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Abstract

Most macroeconomic models impose a tight link between the term structure of interest rates and expected future short rates via the expectations hypothesis (EH). While systematically rejected in the data, existing tests of the EH typically assume full information rational expectations, stationarity of beliefs, or both. As such, they are ill-equipped to refute the EH when these assumptions fail to hold, leaving the door open for a “resurrection.” We re-evaluate the EH using direct measures of expected short rates from all available U.S. surveys of professional forecasters, combined with a parsimonious model of expectations formation to construct the term structure of expectations. We show that deviations from rationality and time variation in long-run beliefs consistent with the observed survey data, while sizable, do not come close to bridging the gap between the term structure of expectations and the term structure of interest rates. We introduce a novel test of the EH and show that the EH is decisively rejected in the data. Outside of short maturities, expectations display, at best, only a weak co-movement with the forward rates of corresponding maturities, both unconditionally and in response to a monetary policy shock.

JEL classification: D84, E43, G12

Key words: expectations formation, survey forecasts, expectations hypothesis

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This paper presents preliminary findings and is being distributed to economists and other interested readers solely to stimulate discussion and elicit comments. The views expressed in this paper are those of the author(s) and do not necessarily reflect the position of the Federal Reserve Bank of New York or the Federal Reserve System. Any errors or omissions are the responsibility of the author(s).

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1 Introduction

The expectations hypothesis of the term structure of interest rates states that yields on government bonds reflect the average short-term interest rate expected to prevail over the life of the bond. Since the 1930s, the expectations hypothesis (EH) has been the natural starting point for linking longer-term yields to short-term yields (e.g., [Lutz 1940](#)). To this day, it remains a maintained assumption in most macroeconomic models, where the monetary transmission mechanism relies on a tight relation between expectations of future short rates and longer-term interest rates (e.g., [Woodford 2003](#)). The preeminent role of the EH stands in tension with the overwhelming empirical evidence stacked against it.

However, analyses of the EH are only as good as the expectations formation process they are based on. [Friedman \(1979\)](#) and [Froot \(1989\)](#) first emphasized that standard tests of the EH are, in fact, *joint* tests of the EH and full information rational expectations (FIRE). [Campbell and Shiller \(1991\)](#) find that when the term spread is high, long-term rates tend to fall, rather than rise, as the theory would suggest. However, these patterns are consistent with the EH under bounded rationality and learning ([Sinha 2016](#); [Farmer, Nakamura, and Steinsson 2023](#)). Furthermore, economic agents are commonly assumed to operate in a stationary environment, where they quickly come to understand the long-run behavior of the economy. Their corresponding long-run expectations, therefore, are stable and, under the EH, would fail to match the substantial variability in realized longer-maturity interest rates. Conversely, models where agents constantly shift their views about the monetary regime (e.g., central bank objectives) can produce long-term rates consistent with the EH ([Fuhrer 1996](#); [Kozicki and Tinsley 2001](#); [Gürkaynak, Sack, and Swanson 2005](#)).¹

In this paper we evaluate the EH using survey measures of consensus expected nominal short-term interest rates (short rates) from a unique dataset that covers *all* surveys of professional forecasters in the U.S. Relying on survey data circumvents the need to make specific assumptions about the degree of rationality embedded in expectations or their inherent stationarity. Ideally, we would observe short rate expectations at all time periods and for all horizons from a single source. In practice, no matter the source, survey data feature substantial data gaps across time and forecast horizons. We introduce a parsimonious model that harmonizes these survey data into a full term structure of expectations. Importantly, it allows for, but does not require, deviations from rationality and time-varying long-run beliefs. The model tightly fits these survey responses and generates an entire term structure of short rate expectations ranging from the current period to very long run expectations.

¹Observed long-rates display “excess volatility” relative to those predicted by the EH ([Shiller 1979](#); [Campbell and Shiller 1991](#)).

Our main contribution is to show that the tight connection between short-term interest rate expectations and the term structure of interest rates, assumed to hold in EH theory, demonstrably fails to hold in practice. Expected short rates beyond three years display, at best, only a weak co-movement with forward rates of the corresponding maturities. In fact, the correlation between changes in longer-term forward rates and corresponding longer-horizon short rate forecasts converges toward zero as the maturity increases to ten years. Further, this pattern continues to hold *conditional* on specific economic shocks. We show that in response to a monetary policy shock, identified from high-frequency financial data, the expected path of the short rate and the corresponding forward move in opposite directions.

We also introduce a new formal test of the EH which has appealing properties relative to the regression-based approaches which have been commonly employed in the previous literature. We apply the parameter stability test of [Elliott and Müller \(2006\)](#) to the wedge between forward rates and their corresponding short rate expectations. The EH is rejected decisively across maturities implying that this wedge varies over time. Importantly, this test does not require any assumption about the expectations formation mechanism (e.g., it does not require rational expectations).

Our results are not dependent on how we construct the term structure of expectations. As an alternative to our baseline model, we fit our survey data using a dynamic version of the [Nelson and Siegel \(1987\)](#) (NS) model of the yield curve, recently employed in [Aruoba \(2020\)](#) to study the term structure of inflation expectations. This exercise produces essentially identical results. In addition to introducing an alternative model we show, for the maturities which are available, that the EH is rejected when using raw survey data directly.

Our baseline model reflects that professional forecasters make short rate projections jointly with other key macroeconomic variables (e.g., [Andrade et al., 2016](#); [Crump et al., 2025](#)). As a secondary contribution we show that our parsimonious multivariate trend-cycle model of expectations formation, based on information from 627 forecast series for real GDP growth, CPI inflation, and the 3-month Treasury bill, captures the behavior of both nominal *and* real short rate expectations. This result holds for forecast horizons from the very-short term (current quarter) to the long-run (11 years and beyond).

The model also allows us to assess the degree of deviations of survey forecasts from the full rationality benchmark at every forecast horizon. Following the approach of [Coibion and Gorodnichenko \(2015\)](#) we exploit our rich dataset to explore under- and over-reaction of forecasts to new information for both short- *and* long-forecast horizons. While short rate forecasts at short horizons under-react to new information, at longer horizons we find substantial over-reaction. In fact, deviations from rationality and time variation in long-run beliefs are shown to be sizable in the survey data. Consequently, as in [Sinha \(2016\)](#)

and [Farmer, Nakamura, and Steinsson \(2023\)](#), treating data generated from the model as observed yields (i.e., imposing the EH), would nonetheless lead an empirical researcher to frequently reject the EH using standard statistical tests which rely on the assumption of rational expectations.

An important implication of our main result is that allowing for deviations from FIRE and time variation in long-run beliefs does not come close to bridging the gap between the survey-based term structure of expectations and the term structure of interest rates. The flip side of this result is that the *wedge* between observed yields and expected future short-term interest rates captures the vast majority of yield variability at medium and long maturities. In models where agents are risk averse, this wedge represents time-varying compensation for bearing risk. However, we show that this wedge is only partially explained by the underlying factors shaping beliefs about the state of the economy. This implies that any model designed to explain both the term structure of short rate expectations and the term structure of interest rates would need to involve additional drivers. In turn, these drivers represent a channel of first-order importance to better understand the effect of the term structure of interest rates on the aggregate economy that is absent in most macroeconomic models.

Related Literature This paper is related to a rapidly growing literature using survey data to better understand economic agents’ expectations formation process.² While the literature has not settled on a unified framework, a few lessons have emerged: survey forecasts deviate systematically from FIRE predictions, resulting in predictable forecast errors and over- and under-reaction to new information.³ Moreover, professional forecasters’ interest rate expectations, which are formed in real time, behave quite differently compared to forecasts available to an econometrician observing the full sample.⁴ This indicates that agents learn about the economic environment.

Models where agents operate in a non-stationary environment, and have to constantly adapt to structural change, can generate sizable fluctuations in long-horizon expectations. For example, [Fuhrer \(1996\)](#) and [Kozicki and Tinsley \(2001\)](#) show that interest rate expectations consistent with monetary policy regime shifts during the US post-war period can generate movements in yields under the EH that well approximate the degree of variation in observed yields. Similarly, [Cogley \(2005\)](#) proposes a model of the term structure of interest rates based on a VAR with drifting coefficients and finds that the EH cannot be rejected in

²See [Angeletos, Huo, and Sastry \(2020\)](#) and [Eusepi and Preston \(2023\)](#) for recent surveys.

³See, for example, [Eusepi and Preston \(2011\)](#), [Coibion and Gorodnichenko \(2012\)](#), [Bordalo, Gennaioli, Ma, and Shleifer \(2020\)](#), and [Angeletos, Huo, and Sastry \(2020\)](#).

⁴See [Friedman \(1979\)](#), [Piazzesi, Salomao, and Schneider \(2015\)](#), [Cieslak \(2018\)](#), [Singleton \(2021\)](#), and [Eusepi, Giannoni, and Preston \(2024\)](#).

this environment.⁵ More recently, using models of bounded rationality and learning, [Sinha \(2016\)](#) and [Farmer, Nakamura, and Steinsson \(2023\)](#) provide examples where the EH would be rejected in the data even when it holds in the true data generating process. [Sinha \(2016\)](#) explores a New Keynesian model with long-term government bonds and bounded rationality, where agents learn about the long-run evolution of the economy. [Farmer, Nakamura, and Steinsson \(2023\)](#) propose a univariate trend-cycle model for the short-term interest rate. They show that their calibrated model matches existing stylized facts capturing departures from FIRE using short-term survey forecasts with horizons out to four quarters ahead.

Our results stand in stark contrast to the conclusions from this literature. Imperfect information and learning have the potential to deliver EH-implied yields with similar features as observed yields. However, the expected path of short-term interest rates using forward rates as the “market expectation” is far from the expected path of short-term interest rates as measured from professional surveys.

The argument that deviations from full-information rational expectations could invalidate tests of the EH goes back to [Friedman \(1979\)](#) and [Froot \(1989\)](#). While these earlier seminal papers use survey data to explore the validity of the EH without imposing any assumptions about the expectations formation process, limited data hindered their ability to reach clear-cut conclusions. In fact, both papers relied solely on the Goldsmith-Nagan Survey which only provided one-quarter and two-quarter ahead forecasts. [Friedman \(1979\)](#) used forecasts of the short rate to test the EH out only as far as six-months ahead. [Froot \(1989\)](#), instead, employed survey forecasts of longer maturity instruments, such as 30-year mortgage rates, to evaluate the EH beyond one year. However, these financial instruments have different credit, liquidity, and other risks relative to Treasury securities (e.g., pre-payment risk for mortgage rates). Moreover, survey forecasts of longer-maturity instruments do not solely reflect the evolution of the expected short rate over the life of the security and, thus, lead to only an indirect test of the EH. Our analysis expands on these early approaches by directly using short rate expectations from multiple surveys for all available forecast horizons over a much longer sample. We are the first to document deviations from the EH directly using the entire term structure of expectations of the short rate, including the very long run. This is crucial as we show that the failure of the EH is most acute at the three-year horizon and beyond.

Survey expectations of interest rates have also been used in conjunction with no-arbitrage term structure models to analyze the behavior of yields. [Kim and Wright \(2005\)](#) and [Kim](#)

⁵Similarly, [Dewachter, Iania, and Lyrio \(2011\)](#) propose a structural model where agents learn about the inflation target and show that this model, based solely on macroeconomic factors provides a better fit of observed yields than rational expectations models.

and Orphanides (2012) employ survey forecasts of the nominal short rate at a few select horizons to discipline the dynamics of the state variables under the historical measure in small samples. Wright (2011) uses expectations from an affine term structure model and survey-based expectations to study the global decline in yields beginning in the 1980s. Piazzesi, Salomao, and Schneider (2015) use survey forecasts of the short rate, inflation, and of longer-term Treasuries to disentangle subjective (i.e. survey forecasters’) beliefs and objective (i.e. those of a statistician endowed with full-sample information) beliefs. In contrast to these papers, we generate the entire term structure of short-rate expectations without relying on information from yields.

The remainder of this paper is structured as follows. In Section 2, we introduce the main ideas and the notation used throughout the paper. We detail the multiple sources of survey data that we use. We also discuss our model of expectations formation and estimation approach and document the tight fit to the observed term structure of nominal short rate forecasts. In Section 3, we compare the term structure of short rate expectations to the term structure of interest rates, formally test for the EH, and provide our main results. We end this section with a conceptual discussion comparing the average investor and the marginal investor in the bond market. Section 4 concludes. In the Appendix, we provide additional discussion along with further details about the model and a Supplementary Appendix (hereafter, “SA1”) collects additional results and robustness checks.

2 Expectations Hypothesis and Survey Expectations

2.1 The Expectations Hypothesis

The term structure of interest rates is shaped by the behavior of the one-period (month) log yield denoted by i_t . Throughout, we will refer to this as the short-term interest rate or “short rate.”⁶ Under the EH, the n -maturity log zero-coupon yield, $\text{yld}_t(n)$, may be written as

$$\text{yld}_t(n) = \text{yld}_t^{\text{EH}}(n) + \alpha^{(n)}, \quad \text{yld}_t^{\text{EH}}(n) \equiv \frac{1}{n} \sum_{h=1}^n \mathbb{E}_t [i_{t+h-1}]. \quad (2.1)$$

for some fixed constant $\alpha^{(n)}$. In words, the EH implies that, up to a constant, the observed yield, $\text{yld}_t(n)$, equals the average expected nominal short-term interest rate over the following n months. When the EH fails to hold there will be a time-varying wedge between $\text{yld}_t(n)$ and $\text{yld}_t^{\text{EH}}(n)$ rather than the time-invariant constant $\alpha^{(n)}$.

⁶The log yield of a 1-period bond at time t is defined as $i_t = -p_t^{(1)}$, where $p_t^{(1)}$ is the log price of that bond (see, e.g., Campbell, 2017).

To separate longer-term from short-term expectations, we conduct our analyses in terms of *forward rates*, defined as the current yield of an n -period bond maturing in $n + m$ periods:

$$\text{fwd}_t(n, m) = \frac{1}{n}[(n + m) \cdot \text{yld}_t(n + m) - m \cdot \text{yld}_t(m)]. \quad (2.2)$$

Because our empirical model of expectations is estimated at a monthly frequency, we construct annual forward rates as the annual average of monthly forward rates. For example, a 4Y1Y forward (i.e., the one-year rate, four-years ahead) would set $n = 12$ and $m = 48$ in equation (2.2). Expressing the EH in equation (2.1) in terms of forward rates gives

$$\text{fwd}_t(n, m) = \alpha^{(n, m)} + \text{fwd}_t^{\text{EH}}(n, m), \quad \text{fwd}_t^{\text{EH}}(n, m) = \frac{1}{n} \sum_{h=m+1}^{n+m} \mathbb{E}_t[i_{t+h-1}] \quad (2.3)$$

In words, the expectations hypothesis implies that the forward rate $\text{fwd}_t^{\text{EH}}(n, m)$ equals the average expected nominal short-term interest rate over the n months starting m months hence. We can contrast the term structure of interest rates with the corresponding rates implied by the EH. We define the wedge $\mathbf{w}_t(n, m)$ between the two as

$$\mathbf{w}_t(n, m) = \text{fwd}_t(n, m) - \text{fwd}_t^{\text{EH}}(n, m). \quad (2.4)$$

Under the strong-form of the EH, we have that $\mathbf{w}_t(n, m) = 0$ for every t and pair (n, m) . The weak form of the EH, implies that $\mathbf{w}_t(n, m) = \bar{\mathbf{w}}(n, m)$ is constant for every t . This is the form of the EH that plays a key role in standard structural models in macroeconomics. Beyond the exact form of the EH, we also aim to assess the relative importance of the wedge in the evolution of forward rates.

2.2 Survey-Based Expectations

In conventional tests of the EH, bond yields (forwards) are the only observed variables. The assumption of rational expectations induces orthogonality of forecast errors which forges a link between realized and expected short-term rates that is exploited to test the EH (e.g., [Campbell and Shiller, 1991](#)). However, the EH does not require $\mathbb{E}_t[\cdot]$ to be the rational expectations operator. Hence, under these tests, a rejection may not reflect a rejection of the EH, but instead a rejection of the assumption of rational expectations, as shown in [Sinha \(2016\)](#) and [Farmer, Nakamura, and Steinsson \(2023\)](#). Following the seminal work of [Friedman \(1979\)](#) and [Froot \(1989\)](#), our goal is to investigate the EH using “consensus” survey data from professional forecasters as a proxy for the expected short rate. We will

denote survey expectations of the short rate at time $t + h$, made at time t , by $\mathbb{E}_t^s [i_{t+h}]$. This approach does not rely on specific assumptions about the expectations formation process. It produces a measure of the wedge based on survey forecasts (“survey wedge”),

$$\text{sw}_t(n, m) = \text{fwd}_t(n, m) - \text{sfwd}_t^{\text{EH}}(n, m), \quad (2.5)$$

where

$$\text{sfwd}_t^{\text{EH}}(n, m) = \frac{1}{n} \sum_{h=m+1}^{n+m} \mathbb{E}_t^s [i_{t+h-1}]. \quad (2.6)$$

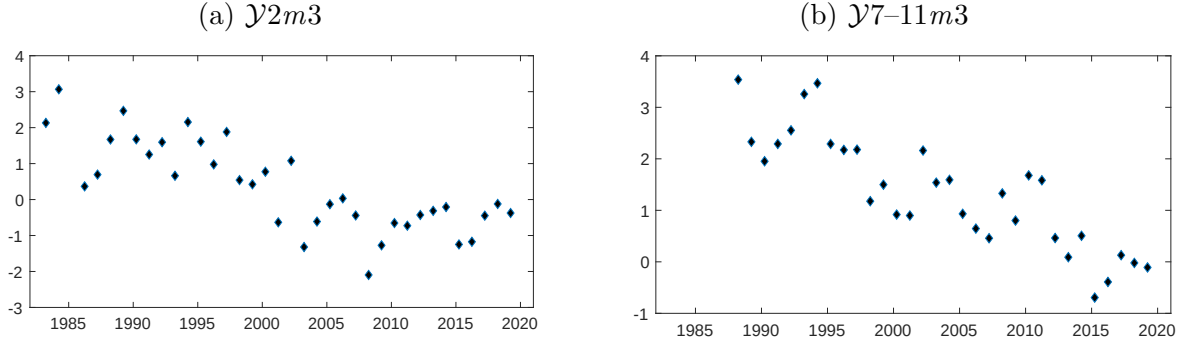
This approach requires observing the full term structure of expectations of the short-term rate at any given time. To this end we employ *all* available surveys of professional forecasters in the United States. There are several advantages of using surveys of professional forecasts instead of surveys of households or firms. First, multiple surveys covering a wide range of forecast horizons spanning “nowcasts” to the very long run are available going back at least to the early 1980s. Second, professional forecasters closely watch the evolution of the economy and the conduct of monetary policy. As they often represent firms active in financial markets, their predictions are likely a good proxy to those of actual traders in the bond market. We provide further discussion of the relation between survey forecasts and market participants in Section 3.6.

Perhaps not surprisingly, available survey-based expectations do not cover all forecast horizons or all time periods. For example, multiple surveys provide responses for the same forecast horizon whereas at other horizons no survey forecasts are observed. Figure 1 shows the survey wedge for two specific survey horizons from the Blue Chip Economic Indicators (BCEI) Survey: the forecast for the 1-year average short rate starting in 21 months, $\mathcal{Y}2m3$, and the five-year average short rate starting in 81 months, $\mathcal{Y}7-11m3$ (see Appendix A for a summary of notation). To obtain the forward rates of the matching horizon, we use the zero-coupon U.S. Treasury yield data from [Gürkaynak et al. \(2007\)](#).⁷ We observe that the survey wedges are large in magnitude and appear to fluctuate over time. However, it is also clear that we have only limited data as these specific forecast horizons are only available, at best, at the annual frequency. This figure provides one example, but is representative of the general limitations of available survey forecasts with forecast horizons beyond one year.

To construct the full path of consensus expectations at all horizons, without unduly over-weighting a particular survey, we introduce a simple parametric model to fit all available survey data. Importantly, the model allows for, but does not require, deviations from ratio-

⁷We use these data to obtain $\text{fwd}_t(n, m)$ for all (n, m) pairs. These data are available at <https://www.federalreserve.gov/pubs/feds/2006/200628/200628abs.html>.

Figure 1: **Examples of the Survey Wedge.** This figure presents the survey wedge, defined in equation (2.5), for selected forward maturities based on the March Blue Chip Economic Indicators Survey. The left chart presents the survey wedge based on a survey horizon of $\mathcal{Y}2m3$ corresponding to a forward maturity of $n = 12$ and $m = 21$ months (see Appendix A for a summary of notation). The right chart presents the survey wedge based on a survey horizon of $\mathcal{Y}7-11m3$ corresponding to a forward maturity of $n = 60$ and $m = 81$ months. The sample period is 1983m3–2019m3 and 1988m3–2019m3, respectively.



nal expectations.

The model serves four purposes. First, by using a model we can provide consistent proxies for the missing observations and/or horizons for each survey and are thus able to extract the common information across different surveys in a coherent way. Second, as professional forecasters make *joint* forecasts across different variables (e.g., [Andrade et al., 2016](#); [Crump et al., 2025](#)), we can utilize the availability of forecasts for output growth and inflation, and exploit the dependence structure across variables and horizons to inform the term structure of forecasts of the short-term interest rate.⁸ This is especially important as we observe fewer forecasts for short-term interest rates than we do for output growth and inflation. Third, the multivariate model allows for the construction of the term structure of real interest rates. This permits the evaluation of the role of expected real rates in shaping the yield curve. Finally, it allows us to assess whether a relatively simple multivariate time-series model can capture the joint dynamics of survey forecasts across the three major macroeconomic variables.

Our dataset combines all available surveys of professional forecasts of the 3-month Treasury bill rate, consumer price index (CPI) inflation, and real GNP/GDP growth at a monthly frequency. Survey data are obtained from nine different sources: (1) Blue Chip Financial Forecasts (BCFF); (2) Blue Chip Economic Indicators (BCEI); (3) Consensus Economics

⁸This is also consistent with a textbook Taylor-rule type reaction function where shifts in short- and long-term views about inflation and economic activity play a key role in the anticipated evolution of the short-term interest rate (e.g., [Andrade et al., 2016](#)).

(CE); (4) Decision Makers' Poll (DMP); (5) Economic Forecasts: A Worldwide Survey (EF)⁹; (6) Goldsmith-Nagan (GN)¹⁰; (7) Livingston Survey (Liv.); (8) Survey of Primary Dealers (SPD); (9) Survey of Professional Forecasters (SPF). For each survey and each forecast horizon we use the consensus forecast, or the mean across forecasters. We focus on three sets of forecasts. For output growth we rely on forecasts of real GNP growth prior to 1992 and forecasts of real GDP growth thereafter. For inflation we use forecasts of growth in the CPI. Finally, we employ the 3-month Treasury bill (secondary market) rate as our measure of a short-term interest rate as it is by far the most frequently surveyed short-term interest rate available.¹¹

Combined, these surveys provide a rich portrait of professional forecasters' macroeconomic expectations. Our results are based on 627 variable-horizon pairs spanning the period from 1983 to 2019. The survey data differ in frequency, forecast timing, target series, sample availability and forecast horizons.¹² Near-term survey forecasts (target period is up to two years ahead) are available for the longest sample with CPI forecasts from the Livingston Survey beginning in the mid-1940s. Medium- and long-term forecasts (target period includes three-years ahead and longer) are available for real output growth and inflation starting in the late 1970s. However, a more comprehensive set of long-term forecasts (a target period of five or more years ahead) for all three variables is available only starting in the mid-1980s which is where we start our sample.

2.3 Filling in the Blanks: A Model of Expectations Formation

Consider an agent who observes the nominal short-term interest rate directly along with noisy signals of other key macroeconomic variables. The agent views the aggregate state of the economy to be defined by $z_t = (g_t, \pi_t, i_t)'$ where g_t is real output growth, π_t is price inflation, and i_t is the short-term nominal interest rate. The perceived law of motion for z_t is

$$z_t = \omega_t + x_t \tag{2.7}$$

$$\omega_t = \omega_{t-1} + \eta_t \tag{2.8}$$

$$x_t = \Phi x_{t-1} + \nu_t \tag{2.9}$$

⁹To our knowledge, the only other paper to use these survey data is [Ehrbeck and Waldmann \(1996\)](#).

¹⁰We thank Kenneth Froot for sharing the Goldsmith-Nagan survey data.

¹¹For example, forecasts of the Federal Funds rate, the target rate of U.S. monetary policy, are only available in two of the nine surveys we consider (BCFF and SPD).

¹²See [Crump, Eusepi, Moench, and Preston \(2023\)](#) for a full discussion of the data sources.

where ω_t is the *trend* component and x_t is a stationary *cycle* component. The innovations, η_t and ν_t , are perceived to be *i.i.d.* Gaussian with joint variance-covariance matrix Σ_z . We assume that the agent perceives the trend component to be slow-moving as compared to the stationary component, implying that the variance of η_t is much smaller than the variance of ν_t .

To form expectations, the agent requires an estimate of the unobserved components x_t and ω_t . However, the agent faces two important informational constraints. First, they observe noisy measures of real output growth and inflation as only the short-term interest rate is fully known at any point in time. This constraint is consistent with the fact that economic variables such as real GDP and inflation are released with a delay and often feature sizable subsequent revisions. In addition, observed measures of price inflation are contaminated with volatile sub-components that mask underlying inflation, π_t , which is the relevant state variable for forecasting.

Second, agents face an additional signal extraction problem. They have to infer to what extent changes in observed data are due to transitory shocks related to the business cycle (innovations to x_t) or reflect shifts in the slow-moving trend components (innovations to ω_t). The latter component may capture regime changes such as long-term shifts in productivity, the savings rate or fiscal policy, which directly impact the evolution of the long term real rate of interest, or shifts in the perceived long-run mean of inflation, reflecting perceived credibility of monetary policy.

We assume that the agent receives two sets of signals, collected in the $m \times 1$ vector $\mathcal{S}_t$. The first set represents the noisy signals of real output growth and inflation coming from economic data releases and the short-term interest rate which is perfectly observed. The agent also observes additional noisy signals on the sub-components, x_t and ω_t . This second set of signals captures information from alternative channels such as other data releases or central bank communications. For example, forward guidance about the short-term path of the policy rate (x_t) or central-bank announcements that alter perceptions about the inflation target (ω_t). The mapping between the agent's model and observed signals is then summarized by the following observation equation

$$\mathcal{S}_t = H' \begin{pmatrix} \omega_t \\ x_t \end{pmatrix} + s_t, \quad (2.10)$$

where H is a $6 \times M$ matrix with M the number of signals and s_t is a vector of measurement errors perceived to be *i.i.d.* Gaussian with variance-covariance matrix Σ_s . The agent uses

the Kalman filter to estimate the latent trend and cycle components,

$$\begin{pmatrix} \omega_{t|t} \\ x_{t|t} \end{pmatrix} = F \begin{pmatrix} \omega_{t-1|t-1} \\ x_{t-1|t-1} \end{pmatrix} + f_t, \quad F = \begin{bmatrix} I_3 & 0 \\ 0 & \Phi \end{bmatrix}, \quad (2.11)$$

where

$$f_t \equiv \mathcal{K} (\mathcal{S}_t - \mathcal{S}_{t|t-1}), \quad \mathcal{S}_{t|t-1} = H' F \begin{pmatrix} \omega_{t-1|t-1} \\ x_{t-1|t-1} \end{pmatrix}. \quad (2.12)$$

In words, we assume the agent observes $\mathcal{S}_t$, posits a multivariate trend-cycle model, and uses the Kalman filter to construct forecasts. The current estimates of x_t and ω_t depend on their previous estimates, along with f_t , which is the “surprise” relative to the agent’s prediction, $(\mathcal{S}_t - \mathcal{S}_{t|t-1})$, scaled by the $6 \times M$ steady-state Kalman matrix $\mathcal{K}$. Given current estimates, $x_{t|t}$ and $\omega_{t|t}$, the agent then generates the *full* term structure of expectations at all forecast horizons, h , via

$$\mathbb{E}_t [z_{t+h}] \equiv z_{t+h|t} = \omega_{t|t} + \Phi^h x_{t|t}. \quad (2.13)$$

Importantly, at any point in time, the term structure of expectations is determined only by $x_{t|t}$, $\omega_{t|t}$ and Φ .

As econometricians, we observe survey forecasts but we cannot directly observe $x_{t|t}$ and $\omega_{t|t}$. In order to estimate $x_{t|t}$ and $\omega_{t|t}$ from the survey data we need to designate their joint dynamic evolution. From equation (2.11), this requires an assumption on the behavior of f_t which we discuss in the next section.

2.4 Subjective Beliefs and the True Law of Motion

If rational expectations were to hold then the agent’s perceived law of motion, as governed by equations (2.7)–(2.9), coincides exactly with the true data generating process of z_t . In this case, f_t is an *i.i.d.* Gaussian process with variance-covariance matrix tied directly to the model’s parameters (see Appendix C for details).

However, the agent’s subjective beliefs need not coincide with the true law of motion of the economy. We would like to allow for the possibility that their statistical model for $\mathcal{S}_t$, as described above, is an approximation (based on incomplete knowledge) of a complex and changing economic environment. Under these conditions, the process f_t is unlikely to satisfy the restrictions imposed by rational expectations. As econometricians, we want to avoid taking a stand on exactly how many or which signals, $\mathcal{S}_t$, the agent observes and their associated true data generating process. Instead, we accommodate possible model mis-

specification and/or deviations from full rationality by allowing the forecast errors, f_t , to be autocorrelated across time. In addition, we do not impose that the variance-covariance matrix of the innovation to the forecast errors satisfies the restrictions imposed by rational expectations. Instead, we assume

$$f_t = Gf_{t-1} + \varepsilon_t, \quad (2.14)$$

where ε_t is *i.i.d.* Gaussian white noise with full variance-covariance matrix Σ_ε and $G = (I_2 \otimes \phi_f)$ with $\phi_f = \text{diag}(\rho_{fg}, \rho_{f\pi}, \rho_{fi})$ where $\otimes$ denotes the Kronecker product.¹³

By equation (2.14), we then have that the forecast errors for the true state of the economy, $z_{t|t} - z_{t|t-1}$, will be serially correlated with first-order autocorrelation coefficients of ρ_{fg} , $\rho_{f\pi}$, and ρ_{fi} , respectively. This modeling assumption aims at capturing the widely documented pattern of serial correlation in forecast errors and under-reaction to shocks in consensus survey-based forecasts (see Section 3.1 below). This pattern is consistent with different (and possibly complementary) information frictions, such as noisy information models (e.g., Coibion and Gorodnichenko 2012; Andrade, Crump, Eusepi, and Moench 2016; Angeletos et al. 2020) and models of imperfect knowledge (e.g., Eusepi and Preston 2011). It is important to emphasize that the dynamic properties of f_t reflect (possible) model misspecification: agents form expectations under the assumption that their model is correct, i.e., that $\mathbb{E}_t[f_{t+1}] = 0$, and so forecasts satisfy equation (2.13), even when G is a nonzero matrix.

The specification in equation (2.14) is chosen for its combination of flexibility and parsimony. We could choose a more general time series process for f_t which could improve the fit of the model further. However, we will show that this choice already fits the data very well and so shifting to alternative specifications would not change any of our conclusions.

In sum, equations (2.11) and (2.14) describe the evolution of the agent's beliefs about the current state of the economy and its sub-components. Given those beliefs, equation (2.13) produces the full term structure of expectations in every time period. While we will show that our model provides a close fit to all of the survey data, the evolution of expectations is regulated by only a small set parameters: $\Theta \equiv (\Phi, \phi_f, \Sigma_\varepsilon)$.

¹³We assume the number of signals M is sufficiently large such that, given the parameters of the model, Σ_ε is nonsingular.

2.5 Bringing the Model to the Data

To estimate the model we use only the professional survey data listed above and do not include any realized economic data.¹⁴ We estimate the parameters of the model, Θ , over the sample period from January 1983 to December 2019. The linearity of the model allows us to use a standard state-space framework. The main challenge is to construct the nonlinear mapping between observed survey forecasts and the parameters implicit in model-implied expectations.

We assume that observed survey forecasts are noisy signals of the true underlying expectations. To cement ideas, suppose our dataset only included monthly forecasts of z_t for different monthly horizons. Let $\text{SVYEXP}_{t+h|t}$ be a 3×1 vector collecting the survey forecasts of real output growth, inflation, and the nominal short rate for the forecast horizon h . Then the observation equation for each survey-horizon pair could be straightforwardly characterized as

$$\text{SVYEXP}_{t+h|t} = \mathbb{E}_t[z_{t+h}] + o_t^{(h)} = \omega_{t|t} + \Phi^h x_{t|t} + o_t^{(h)}, \quad (2.15)$$

where $o_t^{(h)}$ is a horizon-specific measurement error which is assumed to be Gaussian white noise with a diagonal variance-covariance matrix. Even if we were able to observe $\text{SVYEXP}_{t+h|t}$ for every forecast horizon h , equation (2.15) shows that it would still only be a noisy signal of the true expectations.

In reality, we may have multiple surveys for forecasts of the same variable at the same horizon but also face substantial gaps where no forecasts are available. The model allows us to combine these noisy signals in a principled way and fill gaps to obtain model-implied forecasts at all horizons. Furthermore, our observation equation is much more involved than that of equation (2.15). For example, many of the observed survey forecasts are expressed at quarterly or yearly horizons. In addition, recorded forecasts often involve quarterly averages or annualized growth rates. We exploit the fact that all of these more general forecast targets can be well approximated by specific (weighted) linear combinations of $\text{SVYEXP}_{t+h|t}$ for appropriate choices of h which maintains linearity of the observation equation. For example, a forecast formed in the last month of a quarter, for the next quarter's average of the short-term interest rate, may be well approximated by the third element of $\sum_{j=1}^3 \zeta_j \mathbb{E}_t[z_{t+j}] + o_t$ where $\zeta_j = \frac{1}{3}$ for each j . Appendix D provides full details and additional examples.

Overall, the empirical exercise involves a sizable observation equation including 627 time series of survey forecasts for different horizons. To limit the number of parameters to be

¹⁴In theory, we could include the monthly realized 3-month T-bill in the estimation as the agent is assumed to observe it perfectly. In practice, forecasters are surveyed at different times in the month and may not fully observe the average short rate for that month. We verify that the model-implied monthly nowcast of the short-term interest rate is nearly identical to the realized series.

estimated, we group the variances of the measurement errors o_t by the target variable of interest and the horizon of the forecast (but not by the specific survey). In particular we group forecast horizons by: *very short term*, up to two quarters ahead, *short term*, three quarters to two years ahead, *medium term*, between three and four years ahead, and *long term*, five or more years ahead. This allows for a parsimonious yet flexible fitting of observed forecasts.

While the state of the economy is re-estimated in every period, the model’s parameters remain constant, reflecting agents’ invariant priors. This simplifying assumption retains linearity. It is plausible that, over such a long sample, agents would re-evaluate their estimates of the model’s parameters to adapt to structural change. We accommodate this process by allowing for a structural break in the perceived relative variances of the innovations to the trend and cycle components. We date the structural break in the first month of 1999. While the choice of the month may appear arbitrary, it reflects two considerations. First, as shown by [Carvalho, Eusepi, Moench, and Preston \(2023\)](#), long-term inflation expectations begin to stabilize around that period. Separately, [Hanson, Lucca, and Wright \(2021\)](#) show that the sensitivity of long-term interest rates to changes in short term interest rates declined substantially after 1998. This evidence suggests important revisions to agents’ priors about the variability of long-term trends. The structural break allows our model to capture this revision in beliefs using the observed term structure of expectations.

Priors and Posteriors. We use a Bayesian estimation approach which requires prior distributions for the parameters of the model. The variance-covariance matrix of ε_t in equation (2.14) can be expressed as

$$\Sigma_\varepsilon = \begin{bmatrix} \sigma_\omega & 0 \\ 0 & \sigma_x \end{bmatrix} \times C_\varepsilon \times \begin{bmatrix} \sigma_\omega & 0 \\ 0 & \sigma_x \end{bmatrix} \quad (2.16)$$

where $\sigma_x, \sigma_\omega > 0$ are vectors collecting the standard deviations of the innovations to f_t which are linked to the evolution of $\omega_{t|t}$ and $x_{t|t}$ by equation (2.11), and C_ε is a correlation matrix. We assume that the perceived trend component, ω_t , measuring structural shifts in the economy, is slow-moving relative to the cycle component, x_t . This corresponds to the following assumption for our priors. Define the three-dimensional vector measuring the element-by-element ratio $\lambda \equiv \sigma_\omega \odot \sigma_x^{-1}$ where $\odot$ denotes the Hadamard product. We let each element of this vector be distributed according to a beta distribution. As shown in Table 1, the prior mean for all elements in λ is 10% with 90% of the density between 6% and 14%. We maintain the same priors across regimes, as we expect the data to be informative about any difference in perceived beliefs about the long run across these two time periods, resulting in

estimates $\lambda_{1983-1998}$ and $\lambda_{1999-2019}$. Priors on the elements of σ_x are fairly diffuse—set as an independent inverse Gamma distribution with a prior mean of 0.1 and a standard deviation of 2.

The prior on the correlation matrix, C_ε , is defined by the Lewandowski-Kurowicka-Joe (LKJ) distribution (Lewandowski, Kurowicka, and Joe 2009). In terms of moments, the prior mean is the identity matrix I_n with $n = 6$ and the scalar ψ regulates the variance of each off-diagonal entry.¹⁵ We choose a fairly loose prior value on this parameter, setting $\psi = 2$.¹⁶ For reference, $\psi = 1$ implies a uniform distribution across all entries, while $\psi > 1$ favors covariance matrices with stronger diagonal elements (weaker correlations). Turning to the diagonal matrix ϕ_f , each autocorrelation coefficient has a beta distribution with mean 0.2, reflecting a prior with a modest degree of autocorrelation in forecast errors.

Although we observe a large amount of survey data we do not observe any survey forecasts which satisfy equation (2.15) for $h = 1$. The reason is that forecast horizons in surveys of professional forecasters typically follow quarterly or annual increments while our model is estimated at the monthly frequency. This means that every element of our observation equation corresponds to Φ raised to different powers of h which makes full identification of Φ challenging. To circumvent this problem, we impose positivity on the diagonal elements by the choice of a prior which follows a gamma distribution with mean of 0.5 and standard deviation of 0.1.

Table 1 collects summary information on the prior and posterior distribution for the key parameters of our model. Given the wealth of survey information we employ, it is not surprising that the parameters are precisely estimated. In addition, they move substantially away from their prior distribution toward meaningful deviations from FIRE and significant time variation of the long-run mean. The central tendency of the posterior distribution of λ , which is driven by the agent’s belief about the relative variability of the trend component as compared to the cycle, ranges from about 20 to 45 percent before 1999. In contrast, in the second part of the sample the posterior distribution of λ shifts to the left and is centered at values between 8 to 16 percent. Finally, through the lens of our model, the posterior distribution confirms the empirical regularity that survey forecasts exhibit predictable forecast errors, with the central tendency of the autocorrelation coefficients all above 0.5.

¹⁵Let $B(\cdot, \cdot)$ denote the Beta function. The density function is

$$p(C_\varepsilon) = 2^{\sum_{j=1}^{n-1} (2(\psi-1) + n-j)(n-j)} \times \prod_{h=1}^{n-1} (B(\psi + (n-h-1)/2, \psi + (n-h-1)/2))^{n-j} \times \det(C_\varepsilon)^{\psi-1}.$$

¹⁶We also considered a few different choices, but $\psi = 2$ delivered the highest marginal likelihood.

Table 1: **Prior and Posterior Distributions for Selected Parameters.** This table collects information about the prior and posterior distributions for the key parameters of the model. We compute the posterior parameter distribution using the Random Walk Metropolis-Hastings (RWMH) algorithm. See Section SA1-1.1 in SA1 for full details.

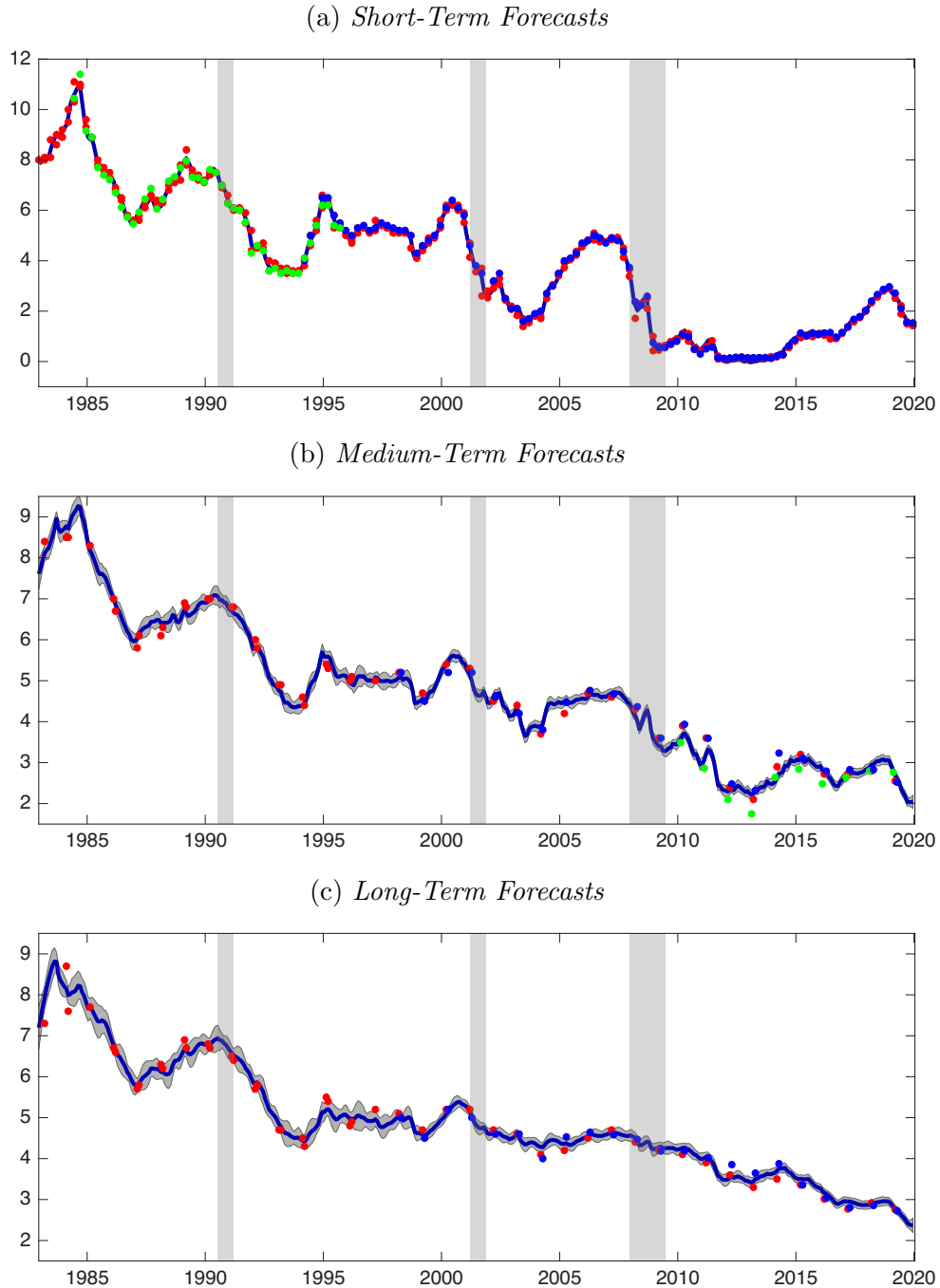
	Prior			Posterior		
	Dist.	Mean	[5% - 95%]	Mode	Mean	[5% - 95%]
$\lambda_{1983-1998}$						
g_t	Beta	0.10	[0.06, 0.14]	0.19	0.19	[0.17, 0.22]
π_t	Beta	0.10	[0.06, 0.14]	0.44	0.44	[0.39, 0.49]
i_t	Beta	0.10	[0.06, 0.14]	0.38	0.39	[0.34, 0.44]
$\lambda_{1999-2019}$						
g_t	Beta	0.10	[0.06, 0.14]	0.08	0.08	[0.07, 0.10]
π_t	Beta	0.10	[0.06, 0.14]	0.14	0.14	[0.12, 0.17]
i_t	Beta	0.10	[0.06, 0.14]	0.16	0.16	[0.13, 0.20]
Forecast Errors						
ρ_{fg}	Beta	0.20	[0.02, 0.49]	0.54	0.55	[0.49, 0.61]
$\rho_{f\pi}$	Beta	0.20	[0.02, 0.49]	0.51	0.52	[0.46, 0.58]
ρ_{fi}	Beta	0.20	[0.02, 0.49]	0.72	0.72	[0.66, 0.77]

The Term Structure of Short Rate Expectations Figure 2 shows the predicted term structure of short rate expectations from our model, together with selected survey forecasts.¹⁷ The model captures the evolution of consensus forecasts of nominal short rates extremely well. The close fit of the model is not a foregone conclusion, given the tight restrictions on the term structure of expectations imposed by the model, as shown in equation (2.13).¹⁸ Given the rich survey data we employ, posterior uncertainty about the model-implied short rate expectations is negligible. In fact, it is visible only for long-horizon expectations for which we have fewer observations.

¹⁷It is important to note that some of the raw survey forecasts have only the approximate horizon relative to the model outputs. Consequently, the fitting errors are, in general, smaller than what is presented in Figure 2. In the second Supplemental Appendix (SA2) available at https://www.dropbox.com/scl/fi/50cul3ezef4l5w7mm96an/CEM_Appendix_Fitted_2026-02-18.pdf?rlkey=8sv64b4nzy9vqekf8ymoud4mn&dl=0 we show the fit for each of the 627 series. To construct the forecast for each horizon we sample the unobserved states ($\omega_{t|t}, x_{t|t}$) using the Carter and Kohn (1994) simulation smoother.

¹⁸The tight fit of our model-implied expectations to the survey data implies that the array of existing stylized facts about survey expectations (e.g., as summarized in Farmer, Nakamura, and Steinsson 2023) are inherited by our measure of expectations.

Figure 2: Term Structure of Short Rate Expectations. This figure compares model-implied survey forecasts to selected individual survey-horizon pairs. The top panel displays four-quarter ahead forecasts taken in the last month of the quarter (“Short-Term”); the middle panel displays forecasts for the one-year average short rate three years ahead taken in the first half of the year (“Medium-Term”); the bottom panel displays forecasts for the one-year average short rate five years ahead taken in the first half of the year (“Long-Term”). The dots indicate forecasts from different surveys: BCEI and BCFF shown in red, CE shown in blue; EF and SPF are shown in green in the top and middle panel, respectively. The black line denotes the median prediction from the model, and the dark grey area the 99% posterior coverage interval. Light grey shaded areas denote NBER recessions. The sample period is 1983m1–2019m12.



The model captures the fact that professional forecasters frequently revise their forecasts at all horizons. Furthermore, interest rate expectations behave as we expect over the monetary policy cycle. Comparing expectations across horizons, at the same point in time, the term structure of policy rate expectations typically flattens (and often inverts) at the end of economic expansions and steepens in the aftermath of recessions. For example, the term structure of short rate expectations inverts in early 1989 when expectations at short-term horizons reached their local peak, leading into the 1990-91 recession. After the short rate reached the zero lower bound in 2008, the term structure of expectations flattened at first, and then steepened again as forecasters persistently expected an imminent lift-off.¹⁹

3 Is There Hope for the Expectations Hypothesis?

3.1 Deviations from FIRE and the EH

As suggested by the estimated parameters, the model replicates frequently reported evidence of deviations from FIRE. Building on this observation, we show that standard tests of the EH based on the assumption of rational expectations can reject the EH even when it holds.

Rejecting FIRE The most common statistic used to illustrate deviations from FIRE is the presence of (first-order) autocorrelation in forecast errors. Table 2 displays the sample autocorrelation of interest rate forecast errors at horizons of one-, two-, three- and four-quarters ahead, from both the raw survey data and those from the fitted model. The forecast errors are computed at a quarterly frequency, to maintain consistency with the empirical results already available in the literature. We calculate autocorrelation coefficients using non-overlapping samples when the forecast horizon is two-quarters ahead or more. For the raw survey data, the table reports results for the SPF and BCEI. While the model is fitted to all nine surveys, we focus on these surveys for two reasons. First, they are frequently used in the literature and have the longest available sample. Second, SPF and BCEI are sampled at different months in the quarter. This allows us to evaluate whether the pattern of forecast errors depends on the specific month of collection, as would be expected.

Table 2 reveals that the fitted model can replicate three important aspects of the data. First, forecast errors are positively correlated for short-term forecasts, displaying “under-reaction” relative to the FIRE baseline. Second, the degree of autocorrelation declines with the forecast horizon. Third, the degree of autocorrelation declines with the month of the

¹⁹Note that these estimated measures of short rate expectations based on survey forecasts are consistent with a perceived zero lower bound (ZLB) on nominal interest rates.

Table 2: **Sample First-Order Autocorrelation of Forecast Errors.** This table reports the sample estimate of the first-order autocorrelation coefficient of quarterly forecast errors of the short-term interest rate. Minimum and maximum statistics are taken across 1,000 posterior draws. The sample period is 1983m1–2019m12.

	1st Month of Quarter				2nd Month of Quarter					3rd Month of Quarter			
	Survey	Model			Survey	Model				Survey	Model		
	BCEI	Min.	Med.	Max.	BCEI	SPF	Min.	Med.	Max.	BCEI	Min.	Med.	Max.
Q1 Ahead	0.496	0.460	0.486	0.512	0.442	0.486	0.397	0.425	0.449	0.353	0.235	0.278	0.323
Q2 Ahead	0.369	0.310	0.329	0.349	0.289	0.280	0.260	0.280	0.304	0.263	0.207	0.230	0.251
Q3 Ahead	0.363	0.293	0.309	0.333	0.252	0.190	0.184	0.201	0.218	0.219	0.150	0.150	0.150
Q4 Ahead	0.200	0.174	0.174	0.174	0.150	0.094	0.097	0.115	0.130	0.118	0.089	0.103	0.124

quarter in which the forecast is queried. This stands to reason as forecasts earlier in the quarter are based on less information than those later in the quarter.

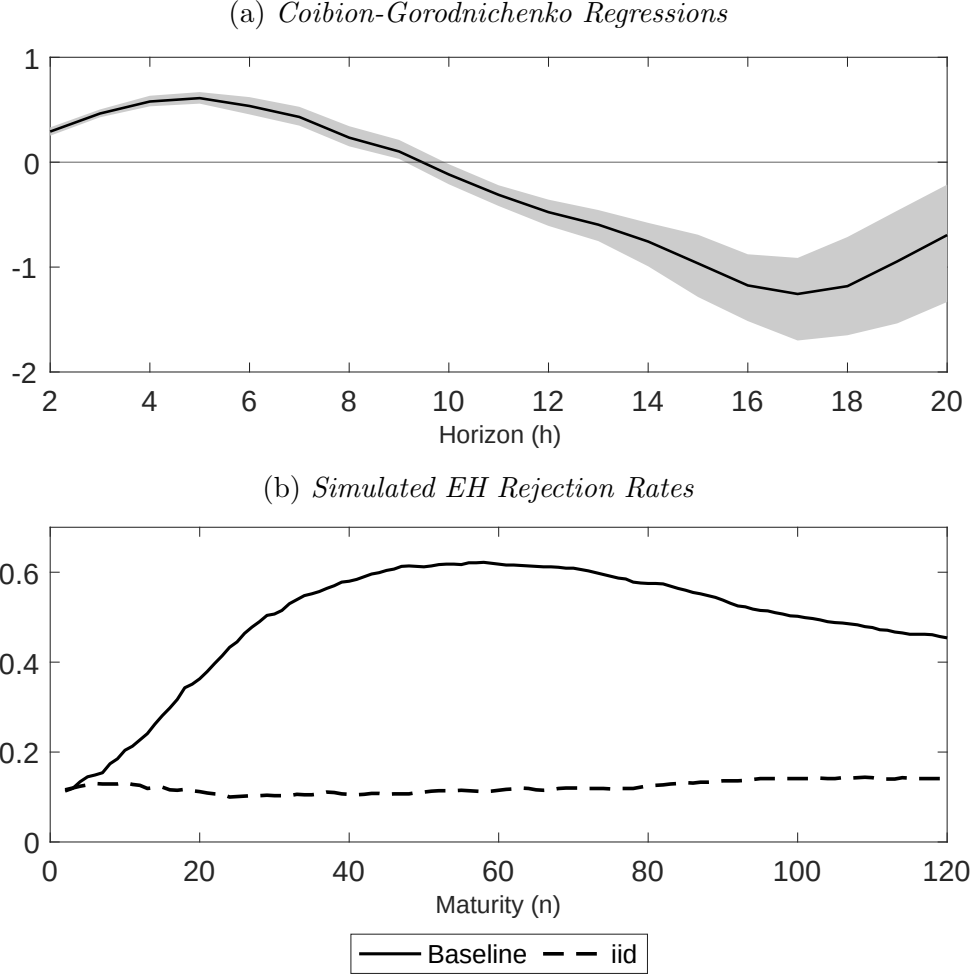
A number of papers also study whether forecasts “over-” or “under-react” to new information relative to the FIRE benchmark. This is commonly measured as the estimated slope coefficient in regressions of the form (see [Coibion and Gorodnichenko 2015](#)):

$$f_{t+h|t}^Q = \alpha_f^{(h)} + \beta_f^{(h)} (\mathbb{E}_t[i_{t+h}] - \mathbb{E}_{t-1}[i_{t+h+1}]) + \xi_{t+h}^{(h)}. \quad (3.1)$$

Here, $f_{t+h|t}^Q$ is the quarterly h -period ahead forecast error from the estimated model and the regressor represents the one-period revision in the h -quarter ahead forecast also from the estimated model. We use the fitted model to estimate $\beta_f^{(h)}$ for horizons from one-quarter ahead to five-years ahead. This substantially expands upon the existing literature which has only utilized short-term survey forecasts (up to four quarters ahead).

The top panel of Figure 3 provides the estimates of $\beta_f^{(h)}$ across forecast horizons of $h = 2, \dots, 20$ quarters. Under FIRE, this slope coefficient should be zero. In contrast we find a positive coefficient for short-term forecasts and a negative coefficient at medium and long-run forecast horizons. In the univariate framework which is common in the literature, this would be consistent with expectations adjusting more slowly, or “under-reacting” to new information, relative to FIRE. For these shorter forecast horizons the regression coefficient increases with the forecast horizon. This is consistent with what is commonly found using survey data directly (see, e.g., [Coibion and Gorodnichenko 2015](#) for inflation and [Farmer et al. 2023](#) for the short-term interest rate).

Figure 3: **Properties of Model-Implied Expectations.** The top chart shows the estimated slope coefficient from the regression in equation (3.1) for $h \in \{2, \dots, 20\}$ quarters. The grey shaded region denotes the minimum and maximum estimated slope coefficients across 1,000 posterior draws. The bottom chart of this figure reports empirical rejection rates of the null hypothesis $\mathbb{H}_0 : \beta^{(s,n)} = 1$, across $s = 1, \dots, 1000$ independent simulations using our baseline model (labelled “baseline”) and under the assumption that forecast errors are not serially correlated (labelled “i.i.d.”). Hypothesis tests are formed using a t-statistic with the equal-weighted cosine variance estimator of Müller (2004) and Lazarus et al. (2018).



The figure shows that the regression coefficient flips for forecast horizons farther than two years in the future. This is consistent with “over-reaction” of longer-run expectations to new information. This pattern is consistent with bounded rationality in long-term expectations (Eusepi et al. 2024) and behavioral theories of expectations formation (Bordalo et al. 2019; Bordalo et al. 2020). This is also consistent with the experimental evidence presented in Afrouzi et al. (2023).

One caveat to the interpretation of these results is that our underlying model is multivariate.²⁰ As such, interpreting these coefficients in terms of “over-” and “under-reaction” to

²⁰For further discussion and evidence in favor of multivariate models of expectation formation using consensus and individual forecasts, see Crump, Eusepi, Moench, and Preston (2025).

new information is not clear cut. Instead we view this exercise as a set of empirical moments that any theory of expectations formation should replicate. These results indicate a clear violation of FIRE across horizons and that expectations at longer horizons over-react which is a form of “excess volatility” relative to the rational expectations benchmark.

Testing the EH when FIRE fails to hold We consider a simulation exercise in the spirit of [Sinha \(2016\)](#) and [Farmer, Nakamura, and Steinsson \(2023\)](#) to show that our term structure of expectations deviates sufficiently from the FIRE benchmark so that the EH is rejected by tests which rely on rational expectations, even when the EH holds. As in [Sinha \(2016\)](#) and [Farmer, Nakamura, and Steinsson \(2023\)](#), we utilize the regression-based test of [Campbell and Shiller \(1991\)](#). We simulate EH-consistent yields using our fitted model parameters and show that rejections of the EH are frequent even though it holds in our simulated data by construction.

The strong form of the EH states that the yield of maturity n is equal to the average expected short rate over the life of the bond. Under the EH, longer-term forward rates are then determined as in equation (2.3). We simulate interest-rate forecasts using equations (2.11), (2.14), and (2.15), with the parameter values set to their posterior mode, and the measurement error, o_t , set to zero for all forecast horizons. For each simulation, s , we obtain $i_t^{(s)}$ along with the associated expectations, $\mathbb{E}_t[i_{t+h}^{(s)}]$, for all t and $h \in 1, \dots, 120$ months. We impose the EH by constructing the simulated term structure of interest rates using equation (2.1) with $\alpha^{(n)} = 0$ for all maturities. This gives $\{(i_t^{(s)}, \text{yld}_t^{(s)}(2), \dots, \text{yld}_t^{(s)}(120)) : t = 1, \dots, T\}$ where T is the same length as our data sample. For each simulation, s , we can test the null hypothesis that the EH holds (i.e., $\beta^{(s,n)} = 1$) in the [Campbell and Shiller \(1991\)](#) regression:

$$\text{yld}_{t+1}^{(s)}(n-1) - \text{yld}_t^{(s)}(n) = \alpha^{(s,n)} + \beta^{(s,n)} \left(\frac{1}{n-1} \right) \left(\text{yld}_t^{(s)}(n) - i_t^{(s)} \right) + u_{t+1}^{(s)}, \quad (3.2)$$

for $n = 2, \dots, N$.

The bottom panel of Figure 3 shows the share of rejections across 1,000 independent simulations of the model for all monthly maturities out to ten years. The nominal size of the test is 10%. For maturities beyond two years, the rejection rates are never below 40%, suggesting that an empirical researcher would frequently conclude that the EH was rejected in the data. As a robustness check, we also include the corresponding results when the data are generated without serial correlation in forecast errors (i.e., under the assumption that $G = 0$). In this case the share of rejections across simulations is essentially equivalent to the nominal size of the test. This shows that the rejections we observe are not driven by finite-

sample properties of the test statistic since, under the joint null hypothesis for this test, the empirical size is very close to the nominal size. Instead this shows that it is specifically the deviation from rational expectations which generates the erroneous rejections of the EH.

3.2 Testing the EH With Survey-Based Expectations

In the previous section we showed that observed deviations from FIRE produce misleading results about whether the EH, in fact, holds. In contrast, we can use our measures of consensus expectations across maturities to test the EH without the need to impose assumptions about rationality. To this end, we make direct comparisons between the term structure of forward rates and the term structure of short rate expectations. Figure 4 visualizes the decomposition of forward rates into the survey wedge, $\text{sw}_t(n, m)$, and the expectations component, $\text{sfd}_t^{\text{EH}}(n, m)$ (see equations (2.5) and (2.6)). Here, we take advantage of our rich data set and model of expectations formation to further decompose expected nominal short rates into expected real short rates and expected inflation

$$\text{sfd}_t^{\text{EH}}(n, m) = \frac{1}{n} \sum_{h=m+1}^{n+m} \mathbb{E}_t^s [i_{t+h-1}] = \frac{1}{n} \sum_{h=m+1}^{n+m} \mathbb{E}_t^s [r_{t+h-1}] + \frac{1}{n} \sum_{h=m+1}^{n+m} \mathbb{E}_t^s [\pi_{t+h}], \quad (3.3)$$

where r_t is the ex-ante, one-period real short rate at time t .

Panels (a)–(c) of Figure 4 present these decompositions for nominal Treasury forward rates for the 1Y1Y, 4Y1Y and 9Y1Y forward maturities. Clearly, the wedges are sizable at all maturities and display substantial variation over time suggesting large deviations from the EH. Visual inspection of the figure suggests the expected real rate plays a dominant role at short maturities, but its relative importance decreases at longer horizons. In contrast, inflation expectations exhibit subdued cyclical fluctuations across all maturities. All components display lower variation after 1999, as captured by the structural shift in the second part of the sample. They also display a downward trend in our sample. While inflation expectations stabilize, both expected real rates and wedges further decline in the aftermath of the financial crisis, with the wedges turning negative during this period.²¹

Importantly, at all maturities including the longest maturity, nominal expected average short rates (the sum of expected real rates and inflation) displays substantial time variation.²² This is a necessary condition for the EH to hold, in practice, as realized interest rates at all

²¹Our finding of a secular decline in the wedges is consistent with the evidence in Wright (2011) who uses an affine term structure model to show that term premiums in the U.S. and in other developed countries have experienced sizable and persistent declines between 1990 and mid-2009.

²²Section SA1-1.2 of SA1 shows that long-horizon expected real rates display similar (or more) variation than those based on the time-series models of Laubach and Williams (2003) and Del Negro et al. (2017).

maturities meaningfully vary over time. Finally, panel (d) in Figure 4 compares the model-implied wedge for the 6Y5Y forward interest rate to raw survey data wedges of similar maturity. Even with the maturity mismatch, the differences are hard to discern, showing that the model-based wedge provides a tight approximation to the corresponding wedge based on the raw survey data.

As a robustness check, and to supplement our main results, we also generate fitted expectations using a dynamic version of the Nelson and Siegel (1987) (NS) model (Diebold et al., 2006). In this model, the term structure of short rate expectations is approximated by three dynamic factors capturing level, slope and curvature—see Aruoba (2020) for an application to inflation expectations. We show that this reduced-form model produces results that are extremely similar to those based on our baseline model (see Section SA1-2 of SA1).

Visually inspecting Figure 4 strongly suggests that the deviation of forwards, $\text{fwd}_t(n, m)$, from the survey-based proxy of $\text{fwd}_t^{\text{EH}}(n, m)$, $\text{sfwd}_t^{\text{EH}}(n, m)$, is large and time-varying. We formalize this observation by introducing a novel direct test of the EH. Let $\zeta_t^{(n, m)}$ be the measurement error from the survey data i.e., we assume that

$$\text{sfwd}_t^{\text{EH}}(n, m) = \text{fwd}_t^{\text{EH}}(n, m) + \zeta_t^{(n, m)}. \quad (3.4)$$

Under the EH we have that

$$\text{sw}_t(n, m) = \text{fwd}_t(n, m) - \text{sfwd}_t^{\text{EH}}(n, m) = \alpha^{(n, m)} - \zeta_t^{(n, m)} \quad (3.5)$$

Then, a direct test of the EH may be formulated as a test of the null hypothesis

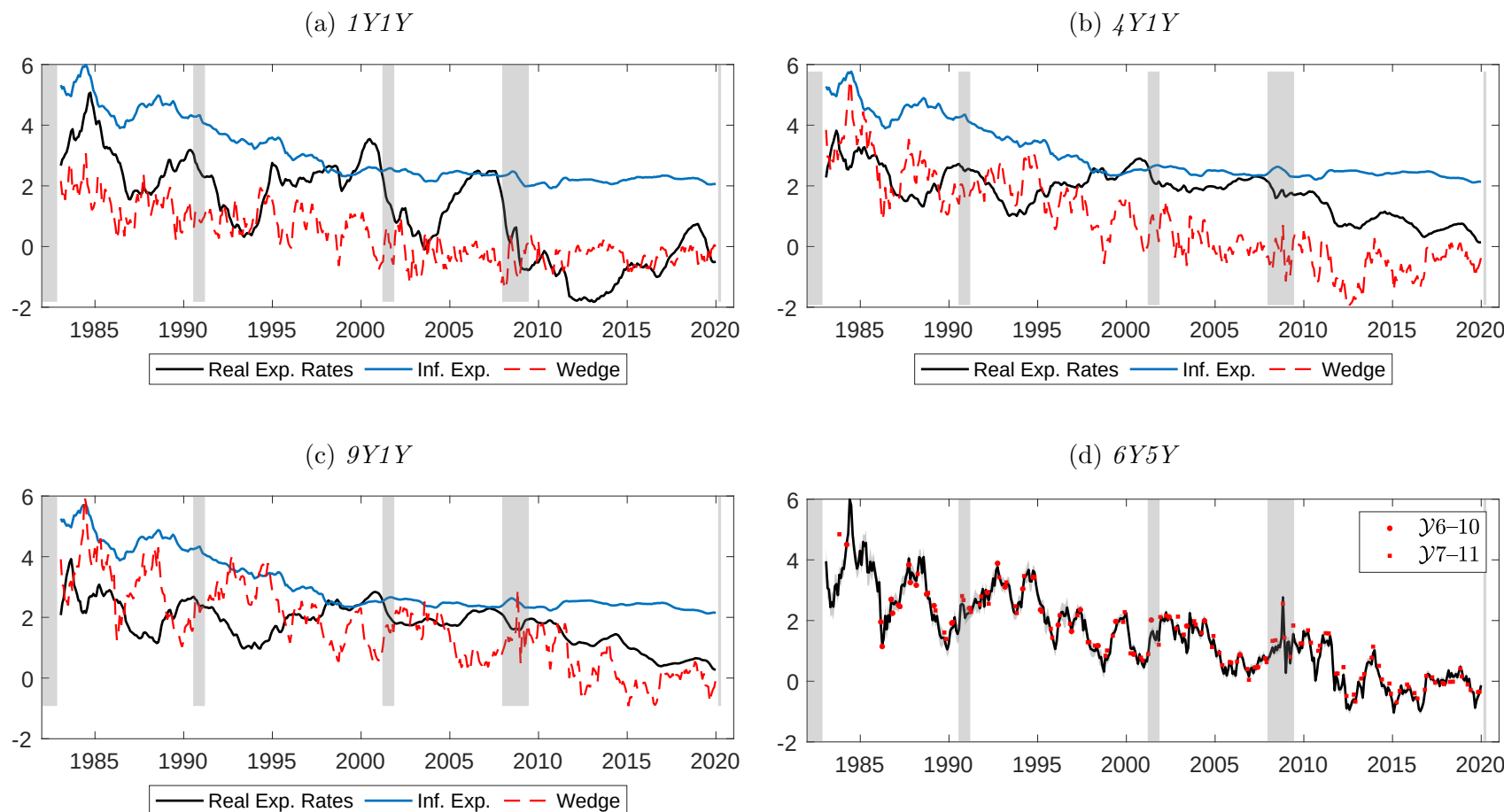
$$\mathbb{H}_0 : \text{sw}_t(n, m) = \bar{\alpha}^{(n, m)} - \zeta_t^{(n, m)} \quad (3.6)$$

for some fixed $\bar{\alpha}^{(n, m)} \in \mathbb{R}$, against the alternative hypothesis

$$\mathbb{H}_A : \text{sw}_t(n, m) = \alpha_t^{(n, m)} - \zeta_t^{(n, m)} \quad (3.7)$$

where $\alpha_t^{(n, m)}$ is contained within a large class of stochastic processes. Intuitively, we are directly testing for parameter stability of $\alpha_t^{(n, m)}$ or, equivalently, whether the wedge, $\text{sw}_t(n, m)$, is constant up to measurement error.

Figure 4: **Decomposing Forward Rates.** The first three charts in the figure show the individual components of forward rates at different maturities. The blue line measures expected inflation, the black line the expected short-term real rate, and the red line the wedge. The forward maturities are 1Y1Y, 4Y1Y and 9Y1Y. The bottom right chart shows the 6Y5Y wedge (black line) with the dark grey area denoting the 99% posterior coverage interval. The dots indicate survey wedges of approximately the same long-horizon maturity using data from BCEI and BCFF. Circles represent survey wedges with forward horizons of $n = 60$ and $m \in \{61, 62, 63, 67, 69, 70\}$ months. Squares represent survey wedges with forward horizons of $n = 60$ and $m \in \{73, 74, 75, 79, 81\}$ months. Light grey areas denote NBER recessions. The sample period is 1983m1–2019m12.



The inference problem described by equations (3.6) and (3.7) was formally studied in Elliott and Müller (2006), who introduce a new test statistic which is asymptotically point-optimal for a broad class of possible processes for $\alpha_t^{(n,m)}$. This class even includes processes with many or clustered structural breaks. A natural assumption on the measurement error, $\zeta_t^{(n,m)}$, is that it is not “too persistent,” or, more formally, that it is weakly dependent which corresponds directly to Condition 3 of Elliott and Müller (2006). Consequently, we construct the test statistic using the Newey and West (1987) variance estimator, as suggested by Elliott and Müller (2006), with the Stock and Watson (2015) data-driven implementation. The simulation experiments in Elliott and Müller (2006) show that the test has good finite-sample properties with empirical size close to nominal size. Finally, it is important to stress that this test does not require an assumption about the expectations formation mechanism and, in particular, we do not require rational expectations.

Table 3: Tests of the Expectations Hypothesis (Model Based). This table reports results of the Elliott and Müller (2006) test applied to the null hypothesis described by equation (3.6) using the model-based short rate expectations for selected forward maturities. For reference, the test rejects for small values of the test statistic with 1% critical value -11.05 and 5% critical value -8.36 . The minimum and maximum values of the test statistic across 1,000 posterior draws are reported. The sample period is 1983m1–2019m12.

	1Y	1Y1Y	2Y1Y	3Y1Y	4Y1Y	5Y1Y	6Y1Y	7Y1Y	8Y1Y	9Y1Y
<i>Full Sample</i>										
Max	-40.5	-47.0	-49.6	-50.2	-49.7	-48.7	-47.5	-46.3	-45.2	-44.6
Min	-42.0	-47.8	-50.4	-51.1	-51.0	-50.3	-49.3	-48.2	-47.2	-46.6
<i>1983–1998</i>										
Max	-17.1	-18.6	-20.9	-21.5	-20.9	-19.9	-18.8	-17.8	-16.9	-16.3
Min	-18.8	-20.3	-22.8	-23.7	-23.6	-22.9	-21.9	-21.0	-20.1	-19.5
<i>1999–2019</i>										
Max	-9.4	-12.9	-18.8	-22.7	-24.5	-25.3	-25.6	-25.9	-26.1	-26.2
Min	-11.1	-14.4	-20.6	-24.5	-26.2	-26.9	-27.2	-27.4	-27.7	-27.9

We implement the Elliott and Müller (2006) test using the short rate expectations estimated from our baseline model.²³ Table 3 reports the test statistics corresponding to one-year forward rates at different forward maturities. We have shown that the model provides a tight fit to the term structure of expectations so that fitting error will not contaminate the results. Nonetheless, we report the minimum and maximum value of the test statistics across 1,000 posterior draws for completeness. The limiting distribution of the Elliott and Müller

²³See page 914 of Elliott and Müller (2006) for step-by-step instructions for constructing the test statistic.

(2006) test statistic is nonstandard and critical values for conventional significance levels are presented in Table 1 of the paper. The test rejects for small values of the test statistic and the 5% and 1% critical values are -8.36 and -11.05 , respectively. The top panel of Table 3 reports results for the full sample. The null hypothesis of a constant wedge is overwhelmingly rejected (p -values all below 1%) across all maturities and posterior draws. This continues to be the case in the middle panel which reports results for the 1983–1998 subsample. The bottom panel reports results for the 1999–2019 sample. For forward maturities beyond one year, we again reject the null hypothesis of a constant wedge with p -values below the 1% level. At the one-year maturity the maximum test statistic is -9.4 and so we comfortably reject the EH at the 5% level. In sum, the test decisively rejects the EH (i.e., that the wedge is constant) across all maturities and subsamples of the data.

As a robustness check, Table 4 reports results using raw survey data. Although we have collected a rich dataset of survey forecasts, only a small subset of the data are candidates for this application because we require a long time series with a consistent maturity structure (for further details, see Appendix B). To do so we rely on monthly forecasts for one to five quarters ahead averaged across multiple surveys as the short-term forecasts. To include medium and longer-term forecasts we use the “Long-Range Consensus U.S. Economic Projections” from BCEI which report short rate expectations two-years ahead (Y2) through six-years ahead (Y6) along with the five-year average rate starting in six years (Y7-11). This is the longest time series of longer horizon forecasts available.²⁴ Even so, these series are only consistently available starting in 1988 and so we only report results for the full sample which represent 63 observations.

The results in Table 4 mirror those of Table 3. For the full sample and the 1983–1998 subsample we reject the null hypothesis of a constant wedge with p -values below the 1% level. For the later subsample, we reject the EH with p -values that are all below 5%. Thus, Table 4 confirms that the results using the model-based forecasts are representative of those using the raw survey data.

How does our new test compare to existing results? In the extant literature, the empirical approach to testing the EH pioneered by Friedman (1979) and Froot (1989) involve regressions of survey expectations on the associated forward rates of the form,

$$\text{sfwd}_t^{\text{EH}}(n, m) = \bar{\alpha}^{(n, m)} + \beta^{(n, m)} \cdot \text{fwd}_t(n, m) + \vartheta_t^{(n, m)} \quad (3.8)$$

²⁴The BCFF also provides forecasts for medium to longer-horizon maturities; however, the BCFF changed the timing of these questions in 1996, which induces a break in the series, and so we rely on the BCEI data only.

Table 4: **Tests of the Expectations Hypothesis (Raw Survey Data).** This table reports results of the [Elliott and Müller \(2006\)](#) test applied to the null hypothesis described by equation (3.6) using short rate expectations based on raw survey data. Forecasts for $Q1$ through $Q5$ are based on a three-month moving average of the underlying series (to ensure constant maturity) and an additional two lags in the associated HAC estimator. Forecasts for $Y2$ through $Y7$ – 11 are based on a two-period moving average of the underlying series (to ensure constant maturity) and an additional lag in the associated HAC estimator. Further details are available in [Appendix B](#) and [Appendix A](#) provides a summary of notation. For reference the test rejects for small values of the test statistic with 1% critical value -11.05 and 5% critical value -8.36 .

	$Q1$	$Q2$	$Q3$	$Q4$	$Q5$	$Y2$	$Y3$	$Y4$	$Y5$	$Y6$	$Y7$ – 11
<i>Full sample</i>	-30.3	-34.4	-35.6	-36.4	-16.3	-10.9	-11.5	-11.5	-11.4	-11.3	-11.1
<i>1983–1998</i>	-14.7	-14.3	-14.0	-13.8	–	–	–	–	–	–	–
<i>1999–2019</i>	-8.7	-9.7	-10.3	-10.3	-10.2	–	–	–	–	–	–
<i>Total Obs.</i>	442	442	442	442	277	63	63	63	63	63	63

or

$$\text{sfwd}_t^{EH}(n, m) - \text{yld}_t(m) = \bar{\alpha}^{(n,m)} + \beta^{(n,m)} \cdot (\text{fwd}_t(n, m) - \text{yld}_t(m)) + \vartheta_t^{(n,m)} \quad (3.9)$$

respectively, for choices of n and m available in the Goldsmith-Nagan survey data. The weak form of the EH implies that $\beta^{(n,m)} = 1$ in either of these regression specifications. Thus, under the null these two specifications are equivalent. These authors draw inferences on whether the EH holds through a t -test of the restriction $\beta^{(n,m)} = 1$. However, a drawback to this approach is that $\beta^{(n,m)} = 1$ is a necessary but not sufficient condition for the EH to hold. Instead, the null hypothesis of the EH is a joint test that $\beta^{(n,m)} = 1$ and the regression error represents only measurement error, i.e., $\vartheta_t^{(n,m)} = \zeta_t^{(n,m)}$ for all t , n , and m (see p.292 of [Froot, 1989](#)).²⁵ Consequently, there can be data-generating processes where the EH fails, but $\beta^{(n,m)} = 1$, so a regression-based test has no power to discriminate between the null and alternative hypotheses. We provide further details and derivations in [Section SA1-1.3](#) of SA1. The key takeaway is that the new test we propose does not suffer the drawbacks of the regression-based approaches as it directly tests for a constant wedge between observed forwards and the survey data.²⁶

²⁵Although this was noted by [Froot \(1989\)](#), the empirical results therein still relied on t -tests of $\beta^{(n,m)} = 1$.

²⁶An additional benefit of our testing approach is that it is invariant to location shifts since it is a direct function of the difference between forwards and survey expectations. In contrast, regression-based approaches will produce different answers depending on the chosen implementation. For example, even with the same data, the OLS estimate of equation (3.8) will differ from the OLS estimate of equation (3.9), in general, even if the EH holds.

3.3 Why do the Tests Reject the EH?

The previous section showed that the EH is rejected decisively; however, a statistical test is a dichotomous outcome. Instead we can ask by how much does the EH fail? To dissect this finding, we provide a simple variance decomposition of forward rates into expected inflation, expected real rates and the survey wedge using 12-month changes in each variable. We have that

$$\hat{\mathbf{s}}\left(n^{-1} \sum_{h=m+1}^{n+m} \Delta_{\tau} \mathbb{E}_t^{\mathbf{s}}[r_{t+h-1}]\right) + \hat{\mathbf{s}}\left(n^{-1} \sum_{h=m+1}^{n+m} \Delta_{\tau} \mathbb{E}_t^{\mathbf{s}}[\pi_{t+h}]\right) + \hat{\mathbf{s}}(\Delta_{\tau} \mathbf{sw}_t(n, m)) = 1. \quad (3.10)$$

Here,

$$\hat{\mathbf{s}}(\cdot) = \frac{\widehat{\text{Cov}}(\Delta_{\tau} \mathbf{fwd}_t(n, m), \cdot)}{\widehat{\text{Var}}(\Delta_{\tau} \mathbf{fwd}_t(n, m))} \quad (3.11)$$

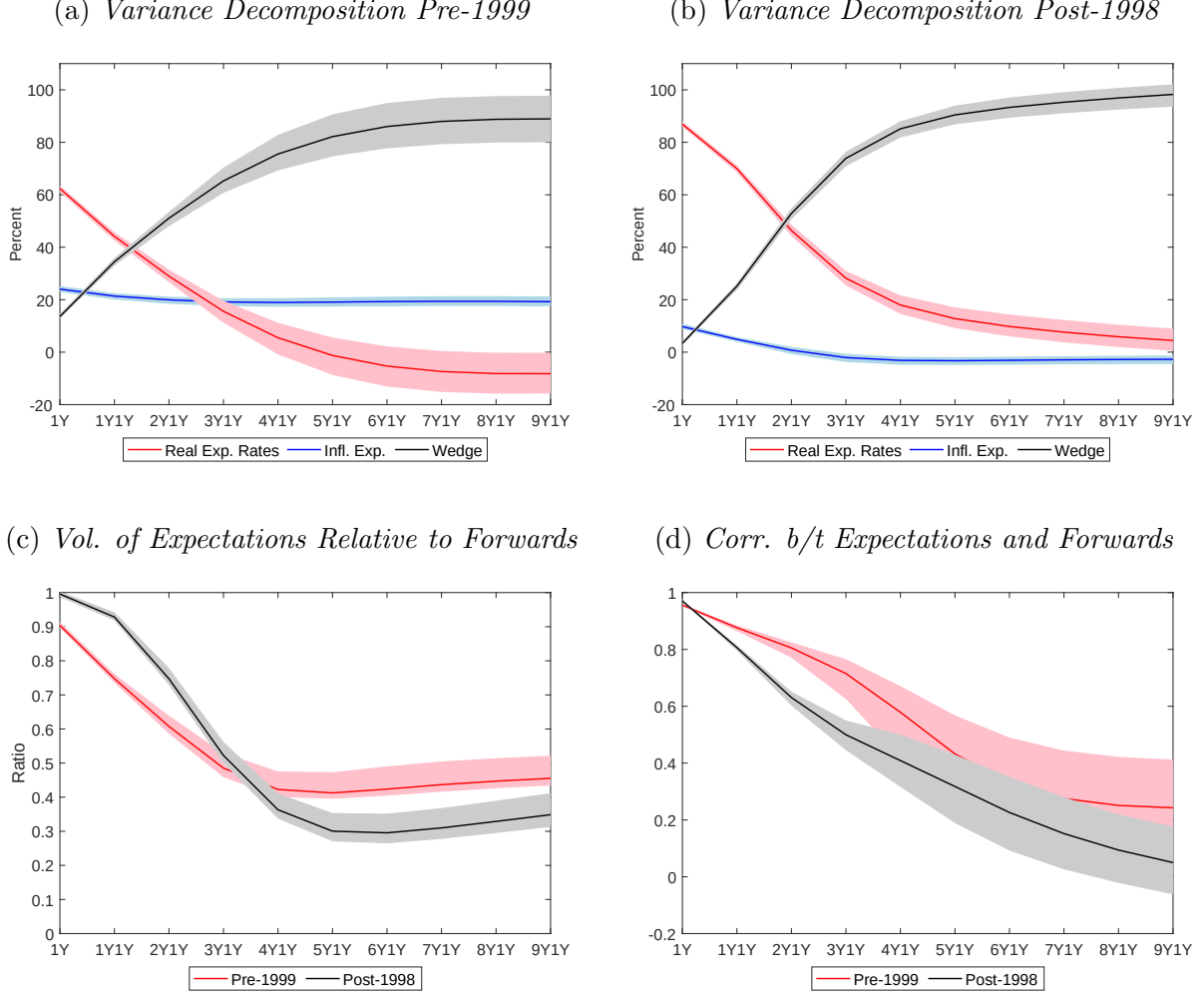
is the ratio between the corresponding sample covariance between forward rates and each constituent component and the sample variance of the forward rate, and $\Delta_{\tau} \varsigma_t = \varsigma_t - \varsigma_{t-\tau}$ for any time series ς_t .

The top panels in Figure 5 show the contributions of the three components to the variance of forwards in the sample before and after 1998.²⁷ The decomposition highlights the striking finding that beyond a maturity of three years (i.e., the 2Y1Y forward), the majority of the variation in forward rates is driven by movements in the wedge rather than short rate expectations. Expected nominal short rates explain between 86% and 97% for the 1-year yield and between 66% and 75% for the 1Y1Y forward rate, and less than 50% for all other forward maturities. As the forward maturity increases, the contribution of expectations to the variability of forward rates declines sharply. This pattern is not affected by the specific sample selected, suggesting that the volatility of forward rates has also declined in the post-1998 period. The only notable difference between sub-samples is the contribution of inflation expectations. In the pre-1999 period, inflation expectations contribute about twenty-percent to the variance of forward rates across all horizons. Conversely, their contribution is close to zero at all horizons in the post-1998 sample. This is mirrored by an increased role of expected real rates at shorter maturities. Finally, each plot also provides shaded regions reflecting sampling uncertainty about the model-generated expectations across 1,000 posterior draws. All the patterns and conclusions are unchanged when considering any of the individual posterior draws.

The bottom panels in Figure 5 offer a further insight into the failure of the EH. The sample

²⁷Note that individual contributions must sum to one but can be negative, depending on the sign of the covariance.

Figure 5: **Decomposing Forward Rates.** The top panels show the variance decomposition of the 12-month changes of forward rates at different maturities in expected real rates (red line), inflation expectations (blue line) and the wedge (black line) components. The bottom left panel displays the volatility of 12-month changes in interest rate expectations relative to forwards for the pre-1999 (red line) and post-1998 (black line) sample. Solid lines denote posterior medians while shaded regions denote the range of 1,000 posterior draws. The sample period is 1983m1–2019m12.



variance share of the components, $\hat{\mathbf{s}}(\cdot)$, can be re-expressed in terms of sample standard deviations and correlations

$$\hat{\mathbf{s}}(\cdot) = \widehat{\text{Corr}}(\Delta_\tau \text{fwd}_t(n, m), \cdot) \left(\frac{\widehat{\text{Var}}(\cdot)}{\widehat{\text{Var}}(\Delta_\tau \text{fwd}_t(n, m))} \right)^{1/2}. \quad (3.12)$$

As we have already shown, the term structure of short-rate expectations implied by our estimated model exhibits substantial volatility. The bottom left panel supports this claim: the volatility of expectations represents a significant portion of the variability of forward rates

across all maturities. This would suggest a more important role for expectations even at the longest horizons. However, as indicated by the bottom right chart, the correlation between forward rates and expectations declines steadily towards zero as the maturity increases.

Importantly, these conclusions do not depend on the frequency of interest rate changes. We choose one-year changes because it eliminates high frequency fluctuations in asset prices that may be short lived. Figure 6 replicates the structure of Figure 5 for the 9Y1Y forward only but now for 1-year, 2-year, 5-year and 10-year changes.²⁸ When we expand the frequency up to five-year changes the conclusions from Figure 5 are essentially unchanged. The survey wedge explains the vast majority of the variation in the forward rates for either the first or second half of the sample. At the same time, the relative volatility of expectations as compared to forwards is higher than for one-year changes. For 10-year changes, which are below business-cycle frequency, the presence of the downward trend in both yields and expectations in the first half of the sample plays a larger role. In the pre-1999 sample, the relative volatility of expectations as compared to the longer-horizon forward is close to one. Despite that, the 10-year change in the 9Y1Y forward expected short rate only explains about half of the variation in the 9Y1Y forward rate. In the second half of our sample, when the downward trend is diminished, the variance decomposition is not significantly altered when calculated with 10-year changes as compared to the higher frequency changes. In fact, the wedge explains over 75% of the variation in longer-horizon forward rates. Taken in sum, not only is the EH decisively rejected in the data, but survey-based short rate expectations, outside of the shortest maturities, display, at best, only a weak co-movement with the forward rates of corresponding maturities.

3.4 Conditional Properties

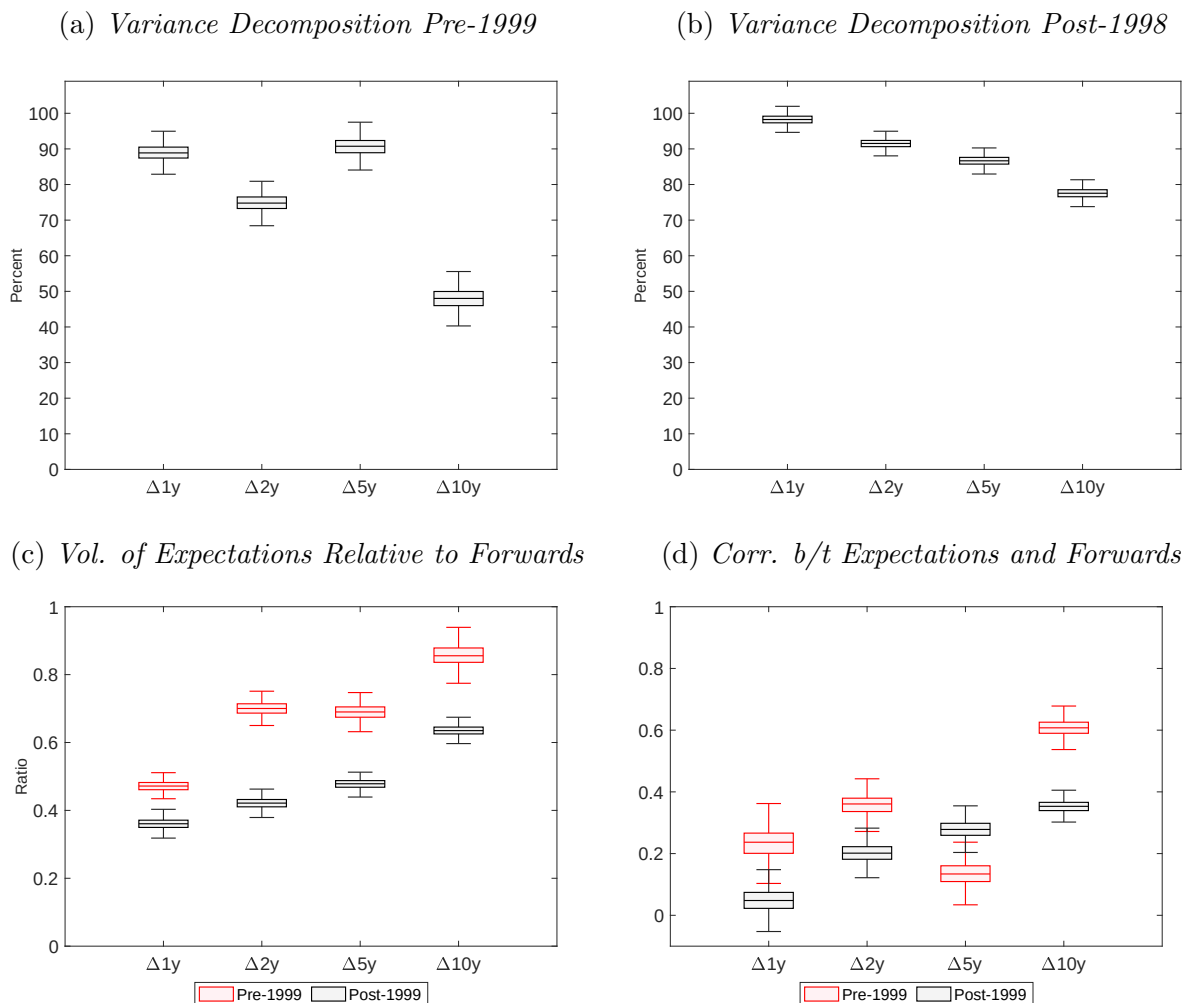
Thus far, we have focused on the unconditional relation between expected short rates and the corresponding forward rates. We have shown that unconditionally, the EH is a poor description of the behavior of interest rates at most maturities. In this section, we investigate whether the EH may provide a good approximation *conditional* on specific macroeconomic shocks. Here we focus on the differential responses to a monetary policy shock of the term structure of short rate expectations as compared to the term structure of forwards.²⁹

As our primary measure of a monetary policy shock, we utilize the orthogonalized shock of Bauer and Swanson (2023). This shock is constructed from high-frequency monetary policy surprises at FOMC announcements, measured using the first principal component from the

²⁸In Section SA1-1.4 of SA1, we present the full counterpart of Figure 5 for 2-year, 5-year and 10-year changes for all forward maturities.

²⁹We thank an anonymous referee for suggesting this exercise.

Figure 6: **Decomposing Long-Horizon Forwards at Different Frequencies.** This figure replicates Figure 5 for the 9Y1Y forward rate for different choices of $\Delta\tau$. We choose τ as 1-year, 2-year, 5-year and 10-year increments. The top row shows the variance share of forwards attributed to the wedge over the pre-1999 and post-1998 samples for different values of τ . The bottom left chart displays the relative volatility of changes in 9Y1Y short-rate expectations compared to the 9Y1Y forward rate for the pre-1999 (red) and post-1998 (black) sample. The bottom right chart shows the correlation between changes in expectations and forward rates for the pre-1999 (red) and post-1998 (black) sample. For all charts, the boxes indicate the interquartile range (with center line equal to the median) and the whiskers represent the maximum and minimum across 1,000 posterior draws. The sample period is 1983m1–2019m12.



first four quarterly Eurodollar futures contracts and is available starting in February 1998. To eliminate predictable components, these surprises are subsequently orthogonalized with respect to macroeconomic and financial data preceding the FOMC announcements.³⁰ To

³⁰In Section SA1-1.6 of SA1 we also report results using the Nakamura and Steinsson (2018) monetary policy shocks which are not orthogonalized (see Bauer and Swanson, 2023, p.103). The results are similar to those presented in Figure 8.

estimate impulse responses we rely on instrumental variables local projection estimates from the following specification in [Bauer and Swanson \(2023\)](#),

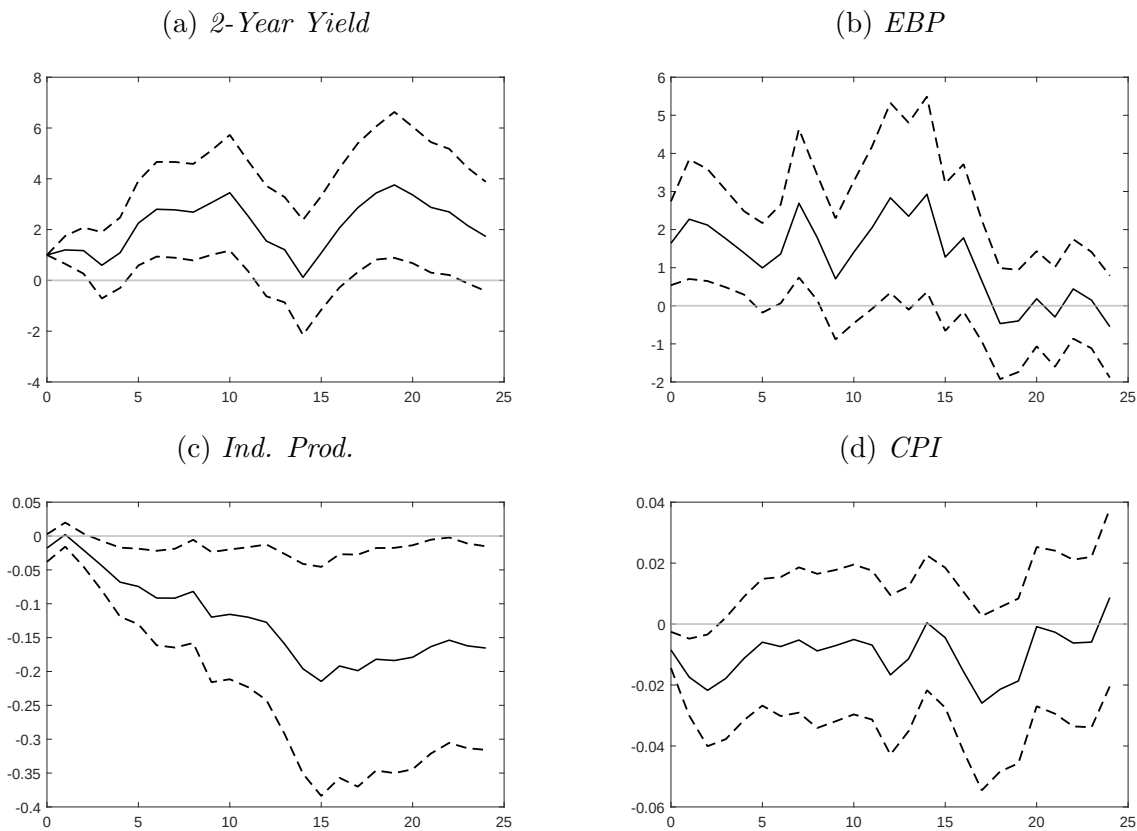
$$Y_{t+h} = a^{(h)} + A^{(h)}(L) \cdot \mathcal{Y}_{t-1} + \theta^{(h)} \cdot \text{yld}_t(24) + \varsigma_t^{(h)}, \quad (3.13)$$

where $\theta^{(h)}$ is the parameter of interest and the matrix polynomial $A^{(h)}(L)$ collects the coefficients associated with 12 lags of the following variables: (1) two-year yield, $\text{yld}_t(24)$; (2) log of industrial production index; (3) log of the CPI; (4) the excess bond premium (EBP); (5) the dependent variable, Y_t . The monetary policy shock is then used as an instrument for $\text{yld}_t(24)$. With the large number of lags, we can rely on Eicker-White standard errors following [Olea and Plagborg-Møller \(2021\)](#).

We start by describing the response of the 2-year yield, activity, inflation, and the excess bond premium to this shock. Figure 7 shows the coefficient estimates from equation (3.13) for these four variables. In response to a surprise monetary policy contraction, both the 2-year yield and the excess bond premium rise within the first two years. At the same time, economic activity as measured by manufacturing output, persistently declines. Inflation is estimated to fall although the decline is modest and statistically insignificant at conventional significance levels.

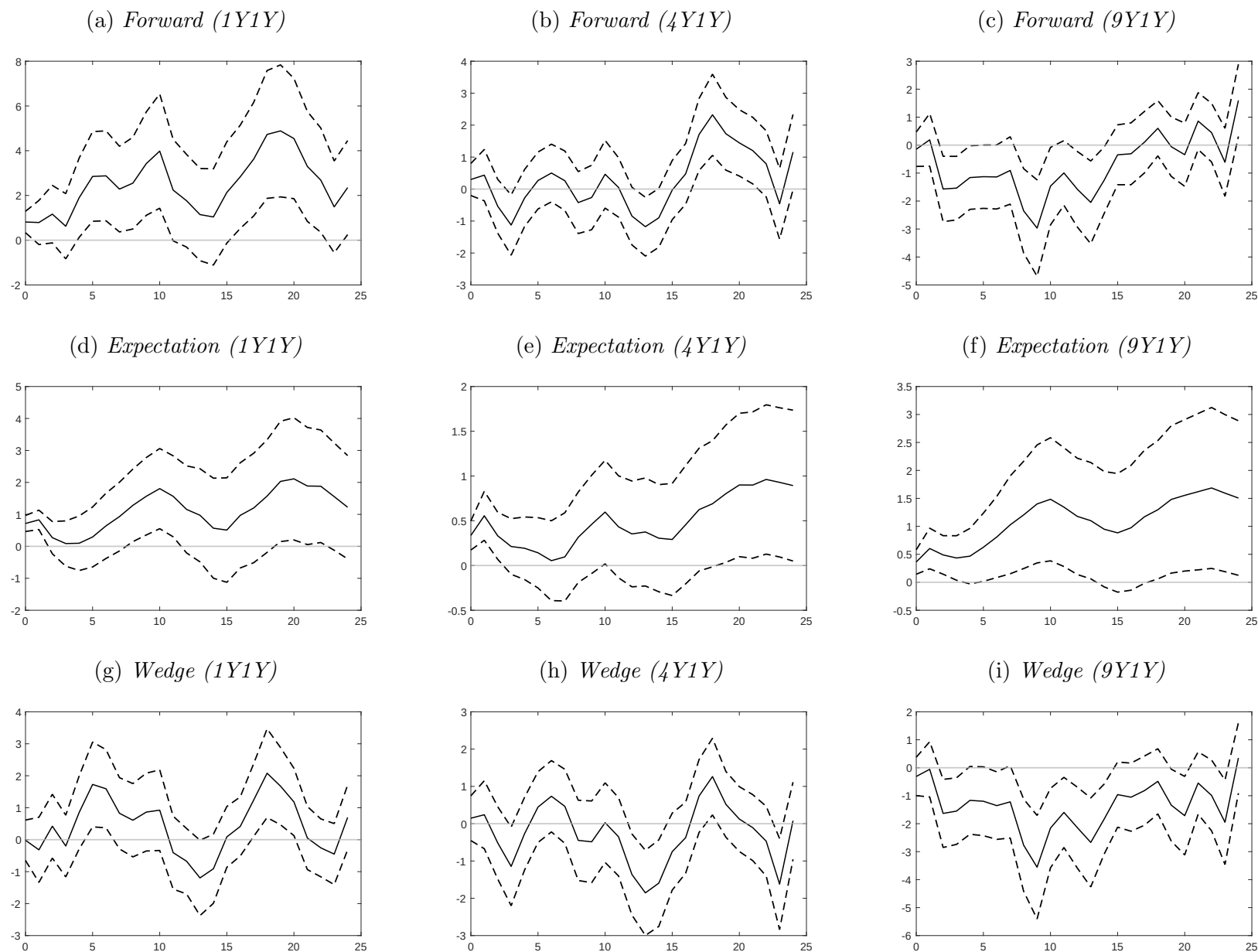
Under the EH, we would expect the conditional response of the term structure of short rate expectations to mimic the behavior of the corresponding forward rates. Said differently, the wedge should show no meaningful response to the shock of interest. Figure 8 demonstrates that this is not the case. The top right chart shows that the 9Y1Y GSW forward rate falls in response to a positive monetary policy shock over the first two years. This echoes the finding in [Gürkaynak et al. \(2005\)](#) that far in the future forward rates decline in response to surprise contractions. However, the right chart in the middle row shows that the corresponding expectations persistently rise in response to the shock: an unexpected tightening of monetary policy leads to an upward revision of short rate expectations. This implies that the long-horizon wedge declines markedly which is shown in the bottom right chart of Figure 8. In the remaining columns of Figure 8, we also show the results for other forward maturities. Similar to the 9Y1Y maturity, the dynamic response of forward rates are aligned with the response of the wedge rather than the response of expectations for the 4Y1Y maturity. At the short maturity, however, expectations and forwards both rise in response to the shock while the response of the wedge varies with time. Thus, the conditional covariance between forwards and expectations is positive at short maturities but diminishes as the maturity grows, consistent with the unconditional evidence presented in Figure 5(d). In sum, these results suggest that, conditional on a monetary policy shock, the EH remains a

Figure 7: **Impulse Response Functions: Macroeconomic Variables.** This figure presents impulse responses to a monetary policy shock of the included macroeconomic variables using the local projection model described by equation (3.13). The coefficient of interest is estimated using instrumental variables with the orthogonalized shock of [Bauer and Swanson \(2023\)](#) as an instrument for the two-year yield. The sample period is 1998m2–2019m12.



poor description of the behavior of the term structure of interest rates, especially at longer maturities.

Figure 8: **Impulse Response Functions: Forwards and Expectations.** This figure presents impulse responses to a monetary policy shock of forwards, expectations, and wedges for selected forward maturities using the local projection model described by equation (3.13). The coefficient of interest is estimated using instrumental variables with the orthogonalized shock of [Bauer and Swanson \(2023\)](#) as an instrument for the two-year yield. The sample period is 1998m2–2019m12.



3.5 Do the Driving Forces of Expectations Explain the Wedge?

Despite the results presented in the previous section, it is possible that the same underlying forces that drive expectations also drive the time variation in the wedge, $w_t(n, m)$. Recall from Equation (2.13) that expectations are linear in six underlying variables, $(\omega'_{t|t}, x'_{t|t})'$, which we filter as $(\hat{\omega}'_{t|t}, \hat{x}'_{t|t})'$. Thus, we can span forecasts at all horizons with these six variables and consider regressions of the form,

$$\Delta_\tau \text{fwd}_t(n, m) = c^{(n, m)} + d_x^{(n, m)'} \Delta_\tau \hat{x}_{t|t} + d_\omega^{(n, m)'} \Delta_\tau \hat{\omega}_{t|t} + \xi_{\text{fwd}, \tau, t}^{(n, m)}, \quad (3.14)$$

and

$$\Delta_\tau \text{sw}_t(n, m) = a^{(n, m)} + b_x^{(n, m)'} \Delta_\tau \hat{x}_{t|t} + b_\omega^{(n, m)'} \Delta_\tau \hat{\omega}_{t|t} + \xi_{\text{sw}, \tau, t}^{(n, m)}. \quad (3.15)$$

We choose τ to be 12, 60 or 120 months. For each of these regressions, we assess the goodness of fit to examine how well forwards and the wedge are spanned by the state variables that drive beliefs about the short-term interest rate.

The top panel of Table 5 reports the regression R^2 for Equation (3.14). Consistent with the evidence we have already presented, the co-movement between forwards and $\hat{x}_{t|t}$ and $\hat{\omega}_{t|t}$ is much stronger at shorter maturities. At medium and longer maturities, the R^2 decline monotonically. A useful benchmark for these results is the class of affine term structure models which explicitly allow for time-variation in the price of risk (e.g., Duffee, 2002; Ang and Piazzesi, 2003; Joslin et al., 2011; Adrian et al., 2013). In this class of models, yields and forward rates are affine functions of the set of risk factors with coefficients obeying specific cross-maturity restrictions. As such, an affine term structure model using $\hat{\omega}_{t|t}$ and $\hat{x}_{t|t}$ as factors can have no higher R^2 than those reported in the top panel of Table 5. Furthermore, if the first three principal components of yields were instead chosen as the risk factors in an affine model—as is the case, e.g., in Joslin, Singleton, and Zhu (2011)—we would observe $R^2 \approx 1$ for all maturities and choice of τ .

We can now investigate directly whether the state variables that drive beliefs also explain the behavior of the survey wedge. The bottom panel of Table 5 reports the regression R^2 for Equation (3.15). When using the wedge as the left-hand side variable, the R^2 values drop precipitously at the short end of the curve and remain low elsewhere. Across all three choices of τ (and across all 1,000 posterior draws), the R^2 is no higher than 50%. In Section SA1-1.5 of SA1, we report the results for a quadratic specification corresponding to all interactions of the six right-hand side variables (27 regressors in all). Although the R^2 values meaningfully increase in this specification, they are still far from fully explaining the

Table 5: **Spanning Tests.** The top panel (bottom panel) presents R^2 from regressions of changes in the forward rate (survey wedge) on a constant and the six state variables, $\hat{x}_{t|t}$ and $\hat{\omega}_{t|t}$. Minimum and maximum statistics are taken across 1,000 posterior draws. The sample period is 1983m1–2019m12.

Forward Rates: $\text{fwd}_t(n, m)$										
	1Y	1Y1Y	2Y1Y	3Y1Y	4Y1Y	5Y1Y	6Y1Y	7Y1Y	8Y1Y	9Y1Y
<i>12-month Changes</i>										
Min. R^2 (%)	92.8	73.6	59.9	49.8	42.7	37.8	34.4	32.3	31.0	30.1
Max. R^2 (%)	93.7	76.5	65.5	58.4	53.4	49.4	46.6	45.0	44.0	43.3
<i>60-month Changes</i>										
Min. R^2 (%)	97.7	90.0	80.5	68.7	57.4	49.2	44.1	40.9	38.7	37.1
Max. R^2 (%)	98.0	90.9	83.0	73.8	65.1	58.5	54.4	51.9	50.0	48.3
<i>120-month Changes</i>										
Min. R^2 (%)	96.4	85.3	76.0	69.0	64.3	61.2	58.9	56.8	54.9	53.3
Max. R^2 (%)	96.9	86.5	79.2	74.3	71.0	68.5	66.4	64.4	62.6	61.1
Survey Wedges: $\text{sw}_t(n, m)$										
	1Y	1Y1Y	2Y1Y	3Y1Y	4Y1Y	5Y1Y	6Y1Y	7Y1Y	8Y1Y	9Y1Y
<i>12-month Changes</i>										
Min. R^2 (%)	7.7	11.8	17.1	20.5	24.1	27.4	29.9	31.7	32.7	33.0
Max. R^2 (%)	14.7	19.5	25.1	30.2	34.8	37.9	39.8	41.0	41.8	42.4
<i>60-month Changes</i>										
Min. R^2 (%)	18.2	20.4	26.3	31.8	34.8	36.2	37.2	38.3	39.5	40.9
Max. R^2 (%)	25.5	26.7	34.1	40.4	44.1	45.6	46.7	47.5	48.3	49.2
<i>120-month Changes</i>										
Min. R^2 (%)	29.0	34.9	38.3	37.2	33.5	29.9	28.3	28.7	29.2	29.0
Max. R^2 (%)	36.2	40.1	44.6	45.4	42.7	39.6	38.1	38.4	38.4	38.1

movements in the wedges. This evidence is strongly suggestive that forces other than those which drive expectations play a key role in explaining the wedge.

To conclude, these results suggest that allowing for time variation in risk premia in otherwise standard macroeconomic models would be insufficient to capture the behavior of the term structure of interest rates, as differences between observed yields and model-implied short rate forecasts are driven by factors separate from those that explain the term structure of short rate expectations.

3.6 Consensus vs. the Marginal Investor

According to the EH, yields reflect “the market” expected path of the short-term interest rate, generally taken to be the average opinion (i.e., the consensus) among market partic-

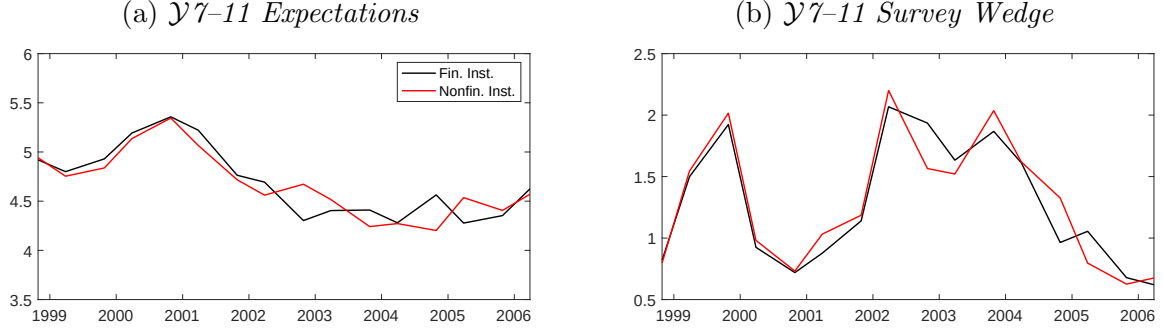
ipants. Following the existing literature, starting from [Friedman \(1979\)](#), we use consensus measures from surveys of professional forecasts as a noisy indicator of the average opinion of the market. The previous sections show that expectations consistent with this consensus measure deviate substantially from observed forward rates. This result is unlikely to stem from severe mis-measurement of the true unobserved average market opinion: professional forecasters are, after all, often themselves market participants. To shed further light on the representativeness of our consensus forecast for market participants’ expectations, we leverage individual long-term forecasts from the Blue Chip Economic Indicators Survey, available over the period 1998–2006 (see also [Crump et al., 2025](#)). This dataset allows us to categorize the survey respondents into “financial firms,” such as Goldman Sachs & Co. or Nomura Securities, and “non-financial firms,” such as the WEFA Group or Ford Motor Company. Figure 9 shows that financial firms have very similar long-horizon forecasts (and corresponding survey wedge) to the non-financial firms.³¹

An alternative criticism is that the “market expectation” does not correspond to that of the average investor but instead to the expectation of the *marginal* investor in the bond market. In this case, our wedge would be attributable to a large and time-varying gap between the beliefs of the average and the marginal investor. Such a large gap is consistent with evidence from individual forecasts in professional survey data. Surveys indicate sizable and time-varying disagreement about the expected path of the short-term interest rate. Importantly, [Andrade, Crump, Eusepi, and Moench \(2016\)](#) show that the term structure of disagreement for the short-rate is upward sloping: belief dispersion increases at longer forecast horizons. This is consistent with the evidence we have presented herein where longer-maturity expectations and the corresponding forwards have little to no relation. Furthermore, [Cao, Crump, Eusepi, and Moench \(2021\)](#) show that professional forecasters’ disagreement about the short-term interest rate co-moves with our survey wedge for long forward maturities, providing empirical evidence for the “disagreement channel.”³²

³¹In unreported results, we also show that this is the case for all other forecast horizons from the BCEI survey.

³²Additionally, [Giacoletti, Laursen, and Singleton \(2021\)](#) show that short-term disagreement in bond yield expectations helps predict bond returns.

Figure 9: **Consensus Forecasts: Financial vs. Non-Financial.** The left chart compares the consensus forecast of the short rate for forecasters at financial firms against those from non-financial firms. The reported series combines the $\mathcal{Y}7-11m3$ and $\mathcal{Y}7-11m10$ forecasts corresponding to $n = 60$ and forward maturities of $m = 81$ and $m = 74$ months, respectively (see Appendix A for a summary of notation). The right chart shows the corresponding survey wedge using these forecasts. Based on individual biannual forecasts from the Blue Chip Economic Indicators Survey. The sample period is 1998m10–2006m3.



These concepts are formalized in a theoretical literature that explores equilibrium bond prices with heterogeneous beliefs. In these models, dispersed beliefs can lead to speculative behavior in the bond market, driving a wedge between bond prices and the EH-implied (consensus) expectations. For example, [Xiong and Yan \(2009\)](#), [Barillas and Nimark \(2017\)](#) and [Buraschi and Whelan \(2022\)](#) show that in standard economies with constant risk aversion and no market frictions, equilibrium bond prices violate the EH and depend on time-varying disagreement. In [Xiong and Yan \(2009\)](#) bond prices are driven by the weighted average of individual investors' expectations, where the weights vary over time and depend on the relative wealth of investors.

In these models, a gap between the average and marginal investor naturally arises. The presence of such a gap is entirely consistent with our finding that the EH does not hold: a key conclusion from this theoretical literature is that, in a market with dispersed beliefs, there does not exist a “market expectation” in the conventional sense. [Jouini and Napp \(2007\)](#) and [Xiong and Yan \(2009\)](#) show that, in a standard complete markets economy, equilibrium asset prices under heterogeneous expectations can be replicated by a “representative investor” with a specific set of beliefs about the expected path of the short rate. However, this representative investor's beliefs do not satisfy standard probability laws, such as the law of iterated expectations. In other words, since the marginal investor changes over time, this representative investor's expectations would not follow an expectations formation process like the one described in Section 2.3. Thus, heterogeneity of beliefs, which would drive a gap between the average and marginal investor's views, in fact, takes us farther away (not closer) from the EH holding, i.e., there is no expectation of “the market” which is consistent

with the term structure of interest rates.

4 Conclusion

In this paper we evaluate the EH using survey measures of consensus expected nominal short rates from a unique dataset that covers all surveys of professional forecasters in the U.S. While short rate expectations move considerably at all horizons and suggest significant departures from rational expectations, they do not come close to matching the observed term structure of interest rates. Instead, the EH-implied short rate expectations generally display, at best, only a weak co-movement with the forward rates of corresponding maturities. Not surprisingly, formal tests of the EH find that it is soundly rejected.

These results suggest alternative explanations for the behavior of observed bond yields such as heterogenous beliefs, financial market frictions, nonstandard risk preferences and behavioral theories of asset pricing. Accommodating such features in models of equilibrium bond prices can have important implications for macroeconomic models, including in the transmission mechanism of monetary policy. In standard models, used by both academics and policymakers, the monetary transmission channel is based solely on the EH. The central bank can exert a tight control on longer-term interest rates by responding to changing economic conditions in a systematic manner, i.e. adhering to time-invariant policy rules, or by communicating directly about likely future policy moves through forward guidance. The sizable deviation of observed interest rates from the EH, which we document, calls into question this conventional framework.

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Appendix

A Notation for Horizons/Maturities

Along with the notation for forward rates introduced in Section 2.1 we also introduce notation to describe survey forecast horizons. $\mathcal{Q}1$ represents a one-quarter ahead forecast, $\mathcal{Q}2$ represents a two-quarter ahead forecast and so on. $\mathcal{Y}1$ represents a one-year ahead forecast, e.g., a forecast for the year 2014 made at any time in 2013. $\mathcal{Y}2$ represents a two-year ahead forecast and so on. $\mathcal{Y}6-10$ represents a forecast for the average value over the years ranging from six years ahead to 10 years ahead, e.g., a forecast for the average short rate from 2020 through 2024, made at any time in 2014. $\mathcal{Y}7-11$ represents a forecast for the average value over the years ranging from seven years ahead to 11 years ahead. The modifier “ $m1$ ” denotes an annual forecast made in January, “ $m2$ ” a forecast made in February, and so on. For example, the 2014 observation of $\mathcal{Y}7-11m3$ is a forecast for the average short rate from 2021 through 2025, made in March 2014.

B Further Details

Table 4: Some care must be taken to use the pure survey data in tests of the EH. Specifically, we need to ensure that there will be a constant wedge when the EH holds. In the case of quarterly forecasts we achieve this by taking a three-month moving average of the data. To see why this is necessary, consider a forecast made in December for the first quarter of the following year. Under the EH, the associated constant wedge would be $(\bar{\alpha}^{(1,1)} + \bar{\alpha}^{(1,2)} + \bar{\alpha}^{(1,3)})/3$. However, a forecast made in November [October] for the first quarter of the following year would be associated with $(\bar{\alpha}^{(1,2)} + \bar{\alpha}^{(1,3)} + \bar{\alpha}^{(1,4)})/3$ $[(\bar{\alpha}^{(1,3)} + \bar{\alpha}^{(1,4)} + \bar{\alpha}^{(1,5)})/3]$. Thus, there could be a time-varying wedge, even if the EH held, purely arising because of the shifting forecast horizon. The three-month moving average eliminates this issue and we can accommodate the transformation by adding two additional lags relative to the data-driven choice in the Newey and West (1987) variance estimator when implementing the test.

A similar issue arises with the BCEI longer-run forecasts. The forecast horizon differs between the March and October editions of the survey. Thus, we transform the semi-annual data using a two-period (one-year) moving average to ensure that, even if the EH held, the wedge would be constant. We then add an additional lag relative to the data-driven choice in the Newey and West (1987) variance estimator when implementing the test.

C Model

The perceived law of motion is as described in the main text. Revisions to the estimates of the state variables evolve as:

$$\begin{pmatrix} \omega_{t|t} \\ x_{t|t} \end{pmatrix} = \begin{pmatrix} \omega_{t|t-1} \\ x_{t|t-1} \end{pmatrix} + \mathcal{K} (\mathcal{S}_t - \mathcal{S}_{t|t-1})$$

where

$$\mathcal{K} \equiv \bar{P}H (H' \bar{P}H + \Sigma_s)^{-1}$$

and $\bar{P}$ solves³³

$$\bar{P} = F \left[\bar{P} - \bar{P}H (H' \bar{P}H + \Sigma_s)^{-1} H' \bar{P} \right] F' + \Sigma_z.$$

Under the correct model (rational expectations) the estimates of the state variables evolve according to

$$\begin{pmatrix} \omega_{t|t} \\ x_{t|t} \end{pmatrix} = \begin{pmatrix} \omega_{t|t-1} \\ x_{t|t-1} \end{pmatrix} + \Sigma_\epsilon^{1/2} \epsilon_t = F \begin{pmatrix} \omega_{t-1|t-1} \\ x_{t-1|t-1} \end{pmatrix} + \Sigma_\epsilon^{1/2} \epsilon_t$$

where the shock vector ϵ_t is normally distributed with zero mean and identity variance-covariance matrix and

$$\Sigma_\epsilon \equiv \mathcal{K} (H' \bar{P}H + \Sigma_s) \mathcal{K}' = \bar{P}H (H' \bar{P}H + \Sigma_s)^{-1} H' \bar{P}.$$

We do not make assumptions about rationality. The agent may have the “wrong” model and departures from rational beliefs can manifest themselves in two ways: First, the agent might have the correct law of motion but the incorrect model parameters, so that their forecast error follows a different distribution. That is, the variance-covariance matrix of the innovations may differ from Σ_ϵ . Second, forecast errors can display serial correlation, as discussed in the main text.

Now we can write the state space of our model as

$$\begin{pmatrix} z_t \\ z_{t-1} \\ z_{t-2} \\ z_{t-3} \\ z_{t-4} \\ \xi_{t|t} \\ f_t \end{pmatrix} = \begin{bmatrix} 0_{3 \times 3} & 0_{3 \times 12} & [I_3 & I_3] F & [I_3 & I_3] G \\ I_3 & 0_{3 \times 12} & 0_{3 \times 6} & 0_{3 \times 6} \\ 0_{3 \times 3} & I_3 & 0_{3 \times 15} & 0_{3 \times 6} \\ 0_{3 \times 6} & I_3 & 0_{3 \times 12} & 0_{3 \times 6} \\ 0_{3 \times 9} & I_3 & 0_{3 \times 9} & 0_{3 \times 6} \\ 0_{6 \times 3} & 0_{3 \times 12} & F & G \\ 0_{6 \times 3} & 0_{6 \times 12} & 0_{6 \times 6} & G \end{bmatrix} \begin{pmatrix} z_{t-1} \\ z_{t-2} \\ z_{t-3} \\ z_{t-4} \\ z_{t-5} \\ \xi_{t-1|t-1} \\ f_{t-1} \end{pmatrix} + \begin{bmatrix} [I_3 & I_3] \\ 0_{3 \times 6} \\ 0_{3 \times 6} \\ 0_{3 \times 6} \\ 0_{3 \times 6} \\ I_6 \\ I_6 \end{bmatrix} \Sigma_\epsilon^{1/2} \epsilon_t$$

where

$$F \equiv \begin{bmatrix} I_3 & 0 \\ 0 & \Phi \end{bmatrix}, \quad G \equiv \begin{bmatrix} \phi_f & 0 \\ 0 & \phi_f \end{bmatrix},$$

and $\xi_{t|t} = (\omega_{t|t}, x_{t|t})'$. Importantly, the measurement equation of the model (from the econo-

³³See Chapter 13 in [Hamilton \(1994\)](#).

metrician's viewpoint) used to fit the survey forecasts is not a function of G . For example,

$$\mathbb{E}_t[z_{t+h}] = e_3 \times \begin{bmatrix} 0_{3 \times 3} & 0_{3 \times 12} & [I_3 & I_3] F & 0_{3 \times 6} \\ I_3 & 0_{3 \times 12} & 0_{3 \times 6} & 0_{3 \times 6} \\ 0_{3 \times 3} & I_3 & 0_{3 \times 15} & 0_{3 \times 6} \\ 0_{3 \times 6} & I_3 & 0_{3 \times 12} & 0_{3 \times 6} \\ 0_{3 \times 9} & I_3 & 0_{3 \times 9} & 0_{3 \times 6} \\ 0_{6 \times 3} & 0_{3 \times 12} & F & 0_{6 \times 6} \\ 0_{6 \times 3} & 0_{6 \times 12} & 0_{6 \times 6} & 0_{6 \times 6} \end{bmatrix}^h \begin{pmatrix} z_t \\ z_{t-1} \\ z_{t-2} \\ z_{t-3} \\ z_{t-4} \\ \xi_{t|t} \\ f_t \end{pmatrix}$$

where e_3 selects the first three rows (see equation (2.15)). Said differently, the agent is assumed to form expectations under the assumption $\mathbb{E}_t[f_{t+h}] = 0$ for all h .

D Measurement equation

In this section we provide greater detail on how we map survey forecasts to our modeling framework discussed in Section 2.5. Forecasts for the three-month Treasury bill rate are either a simple average over a period or end of period. For the latter we assign these forecasts to the last month in the period. For real output growth and inflation, survey forecasts come in three possible forms: quarter-over-quarter (QoQ) annualized growth, annual average growth and Q4/Q4 growth. The distinction between these growth rates is best illustrated through examples. In these examples we will ignore measurement error for simplicity. Let G_{2013Q1} and G_{2013Q2} be the level of real GDP in billions of chained dollars in the first and second quarter of 2013, respectively. Then, QoQ annualized growth rate is defined as $100 \cdot ((G_{2013Q2}/G_{2013Q1})^4 - 1)$. In our model we filter a month-over-month (annualized) real GDP growth rate series. To map the monthly series into this specific quarterly growth rate we follow [Crump et al. \(2014\)](#) and use

$$100 \cdot ((G_{2013Q2}/G_{2013Q1})^4 - 1) \approx \frac{1}{9} (g_{2013m2} + 2 \cdot g_{2013m3} + 3 \cdot g_{2013m4} + 2 \cdot g_{2013m5} + g_{2013m6}),$$

where, for example, g_{2013m2} represents month-over-month annualized real output growth in February 2013.

Annual average growth rates follow a similar pattern. For example, let G_{2012} and G_{2013} be the average level of real GDP in billions of chained dollars in the years 2012 and 2013, respectively. Then the annual average growth rate is $100 \cdot (G_{2013}/G_{2012} - 1)$ which we approximate via,

$$100 \cdot (G_{2013}/G_{2012} - 1) \approx \frac{1}{24} (g_{2012m2} + 2 \cdot g_{2012m3} + 3 \cdot g_{2012m4} + \dots + 12 \cdot g_{2013m1} \\ + 11 \cdot g_{2013m2} + 10 \cdot g_{2013m3} + \dots + 2 \cdot g_{2013m11} + g_{2013m12}).$$

Finally, Q4/Q4 growth rates are calculated, for example, by $100 \cdot (G_{2013Q4}/G_{2012Q4} - 1)$ and approximated via

$$100 \cdot (G_{2013Q4}/G_{2012Q4} - 1) \approx \frac{1}{12} (g_{2013m1} + g_{2013m2} + g_{2013m3} + \dots + g_{2013m12}).$$

The above shows that certain survey forecast horizons will implicitly include time periods which have already occurred. To avoid taking a stand on how forecasters treat past data (e.g., do forecasters use realized data, filtered versions or another measure?) we exclude all survey forecast horizons that include past months' values of y_t . The only exception we make is to include current quarter ($Q0$) and one-quarter ahead ($Q1$) forecasts for real output growth (which extend back, at most, four months and one month, respectively). This is why our measurement equation contains lags of z_t up to z_{t-4} . We do so to help pin down monthly real output growth since the actual series is only available at a quarterly frequency. Finally, for simplicity, forecasts which involve averages over multiple years are mapped as simple averages over the corresponding horizons.

Supplemental Appendix (SA1): “Is There Hope for the Expectations Hypothesis?”

Richard K. Crump, Stefano Eusepi & Emanuel Moench
February 20, 2026

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SA1-1 Additional Details and Results

SA1-1.1 Details on Model Estimation

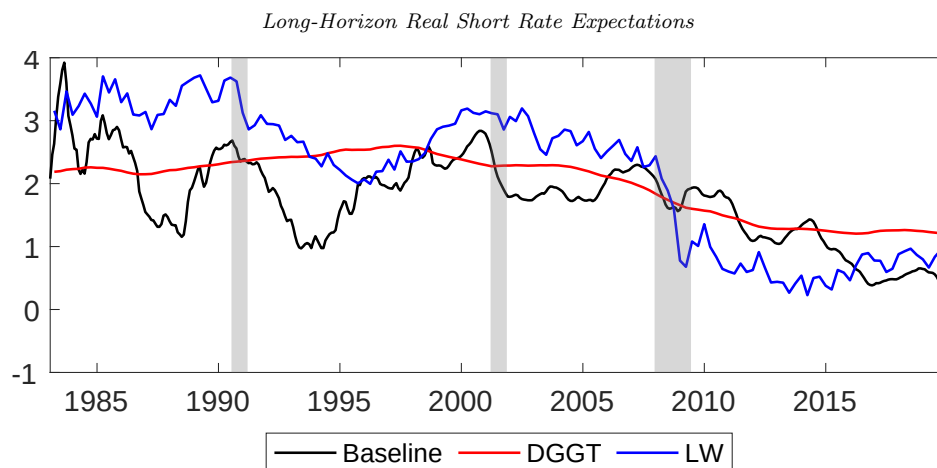
For all model specifications, we compute the posterior parameter distribution using the Random Walk Metropolis-Hastings (RWMH) algorithm. We estimate the mode of the posterior distribution by maximizing the log posterior function. In order to parameterize the proposal distribution (which is assumed to be multivariate normal), we use the Hessian matrix to produce 80,000 draws from the RWMH algorithm. In a second step, we employ the variance-covariance matrix obtained from these initial draws in order to refine the proposal distribution. We then generate 5 chains of 400,000 draws: a step size of 0.3 gave a rejection rate of around 65 percent in each sample (and each model specification). We evaluate convergence using the [Gelman and Rubin \(1992\)](#) potential scale reduction factor (the factor is well below 1.01 for all estimated parameters, in every model specification). The posterior distribution is obtained by combining the 5 chains. The model's posterior coverage intervals are obtained using the [Carter and Kohn \(1994\)](#) simulation smoother. Model predictions and (independent) samples are obtained from 20,000 selected draws from the posterior distribution.

SA1-1.2 Longer-Run Real Short Rate Expectations

Given the variability of observed long-maturity yields, a necessary condition for the EH to hold is that longer-horizon expectations exhibit significant time variation. We illustrate the degree of variability of the model-implied survey forecasts by comparing long-horizon forecasts of the real short term interest rate directly to those from two well-known statistical models. For this exercise, we use real short rates as long-run inflation expectations are roughly constant in the second part of the sample. To compute the model-implied expected real rate, we exploit the multivariate nature of our model and subtract expected inflation from the expected nominal short rate.

As shown in Figure SA1-1, the long-run forecast of the real short rate from our model displays a significant degree of variation when compared to popular alternative model-based measures that explicitly allow for trending interest rates. The red line shows the evolution of the 30-year forecast from the VAR model with time-varying means of [Del Negro, Giannone, Giannoni, and Tambalotti \(2018\)](#) which uses information from both macroeconomic and financial variables. The blue line is the long-run estimate of the short-term real rate from the unobserved components model of [Laubach and Williams \(2003\)](#). We use the one-sided estimate for a better comparison with our survey-based expectations model (black line). The chart shows that our measure of long-run expectations is more volatile than its counterpart in these statistical models. In particular, the standard deviation of twelve-month changes in our model, at 0.43, is higher than 0.35 for [Laubach and Williams \(2003\)](#) and 0.07 for [Del Negro, Giannone, Giannoni, and Tambalotti \(2018\)](#).

Figure SA1-1. Properties of Model-Implied Expectations. This chart displays the model-implied long-horizon forecast of the real short rate using the model introduced in Section 2 of the main text (labelled “Baseline”) as compared to those from [Laubach and Williams \(2003\)](#) (labelled “LW”) and [Del Negro et al. \(2018\)](#) (labelled “DGGT”). The sample period is 1983Q1–2019Q4.



SA1-1.3 Properties of Regression-Based Tests of the EH

In this section we derive the population regression coefficients for the regression specifications in [Friedman \(1979\)](#) and [Froot \(1989\)](#) where, either the left-hand side (LHS) variable is a survey expectation and the right-hand side (RHS) variable is an actual forward, or vice-versa. We illustrate the drawback of solely focusing on the value of β from the regression because $\beta = 1$ is not dispositive about whether EH holds, regardless of the choice of the LHS and RHS variables.

Recall that

$$\mathbf{w}_t(n, m) \equiv \text{fwd}_t(n, m) - \text{fwd}_t^{\text{EH}}(n, m).$$

For simplicity, let us focus on the specification in [Friedman \(1979\)](#) given by equation (2.5). The corresponding population coefficient for the [Froot \(1989\)](#) specification, given by equation (2.6), can be obtained by analogous steps and the qualitative conclusions continue to hold (recall that the regression equations are identical under the EH). We have,

$$\begin{aligned} \beta_F &= \frac{\mathbb{C}(\text{sfwd}_t^{\text{EH}}(n, m), \text{fwd}_t(n, m))}{\mathbb{V}(\text{fwd}_t(n, m))} \\ &= \frac{\mathbb{C}(\text{fwd}_t^{\text{EH}}(n, m) + \zeta_t^{(n, m)}, \text{fwd}_t(n, m))}{\mathbb{V}(\text{fwd}_t(n, m))} \\ &= \frac{\mathbb{C}(\text{fwd}_t(n, m) - \mathbf{w}_t(n, m) + \zeta_t^{(n, m)}, \text{fwd}_t(n, m))}{\mathbb{V}(\text{fwd}_t(n, m))} \\ &= 1 + \frac{\mathbb{C}(\zeta_t^{(n, m)}, \text{fwd}_t(n, m))}{\mathbb{V}(\text{fwd}_t(n, m))} - \frac{\mathbb{C}(\mathbf{w}_t(n, m), \text{fwd}_t(n, m))}{\mathbb{V}(\text{fwd}_t(n, m))}. \end{aligned}$$

If the measurement error is fully independent then the second term is zero. The third term is zero under the null hypothesis but, importantly, can be zero even when the null hypothesis fails so long as $\mathbb{C}(\mathbf{w}_t(n, m), \text{fwd}_t(n, m)) = 0$. This shows that there are settings where a regression-based test can have no power to discriminate between the null and alternative hypotheses.¹

What happens when we reverse the role of the LHS and RHS variables? Then we obtain,

$$\begin{aligned} \beta_{F, \text{rev}} &= \frac{\mathbb{C}(\text{fwd}_t(n, m), \text{sfwd}_t^{\text{EH}}(n, m))}{\mathbb{V}(\text{fwd}_t^{\text{EH}}(n, m) + \zeta_t^{(n, m)})} \\ &= \frac{\mathbb{C}(\text{fwd}_t(n, m), \text{fwd}_t^{\text{EH}}(n, m) + \zeta_t^{(n, m)})}{\mathbb{V}(\text{fwd}_t^{\text{EH}}(n, m) + \zeta_t^{(n, m)})} \\ &= \frac{\mathbb{C}(\text{fwd}_t^{\text{EH}}(n, m) + \zeta_t^{(n, m)} - \zeta_t^{(n, m)} + \mathbf{w}_t(n, m), \text{fwd}_t^{\text{EH}}(n, m) + \zeta_t^{(n, m)})}{\mathbb{V}(\text{fwd}_t^{\text{EH}}(n, m) + \zeta_t^{(n, m)})} \\ &= 1 - \frac{\mathbb{C}(\zeta_t^{(n, m)}, \text{fwd}_t^{\text{EH}}(n, m) + \zeta_t^{(n, m)})}{\mathbb{V}(\text{fwd}_t^{\text{EH}}(n, m) + \zeta_t^{(n, m)})} + \frac{\mathbb{C}(\mathbf{w}_t(n, m), \text{fwd}_t^{\text{EH}}(n, m) + \zeta_t^{(n, m)})}{\mathbb{V}(\text{fwd}_t^{\text{EH}}(n, m) + \zeta_t^{(n, m)})} \end{aligned}$$

¹It is, however, important to note that the sample estimates of $\mathbb{C}(\mathbf{w}_t(n, m), \text{fwd}_t(n, m))$, using $\hat{\mathbb{C}}(\mathbf{sw}_t(n, m), \text{fwd}_t(n, m))$, imply estimated values of $\beta_F^{(n, m)}$ which are much smaller than 1 outside of the shortest maturities. For example, for forward maturities beyond 2Y1Y, the sample analogs of $\beta_F^{(n, m)}$ are all below 0.65.

If the measurement error is fully independent then we have,

$$\beta_{\text{F,rev}} = 1 - \frac{\mathbb{V}(\zeta_t^{(n,m)})}{\mathbb{V}(\text{fwd}_t^{\text{EH}}(n, m) + \zeta_t^{(n,m)})} + \frac{\mathbb{C}(\mathbf{w}_t(n, m), \text{fwd}_t^{\text{EH}}(n, m))}{\mathbb{V}(\text{fwd}_t^{\text{EH}}(n, m) + \zeta_t^{(n,m)})}.$$

Under the EH, $\beta_{\text{F,rev}} \neq 1$ because of the presence of measurement error in the survey data (attenuation bias). Furthermore, when the EH fails, $\beta_{\text{F,rev}}$ may be equal to 1 so long as the latter two terms in the equation sum to zero. Thus, reversing the LHS and RHS variables, unfortunately, still leads to settings where the regression-based test can have no power to discriminate between the null and alternative hypotheses.

SA1-1.4 Variance Decompositions: 2-Year, 5-Year and 10-Year Changes

In this section we supplement the variance decompositions for 1-year changes shown in Figure 5 of the main text. In particular, we provide the corresponding variance decompositions for 2-year, 5-year and 10-year changes in Figures SA1-2, SA1-3, and SA1-4, respectively.

Figure SA1-2. Decomposing Forward Rates (2-Year Changes). The top panels show the variance decomposition of the 24-month changes of forward rates at different maturities in real expected rates (red line), inflation expectations (blue line) and the wedge (black line) components. The bottom left panel displays the volatility of 24-month changes in short rate expectations relative to forwards for the pre-1999 (red line) and post-1998 (black line) sample. Solid lines denote posterior medians while shaded regions denote the range of 1,000 posterior draws. The sample period is 1983m1–2019m12.

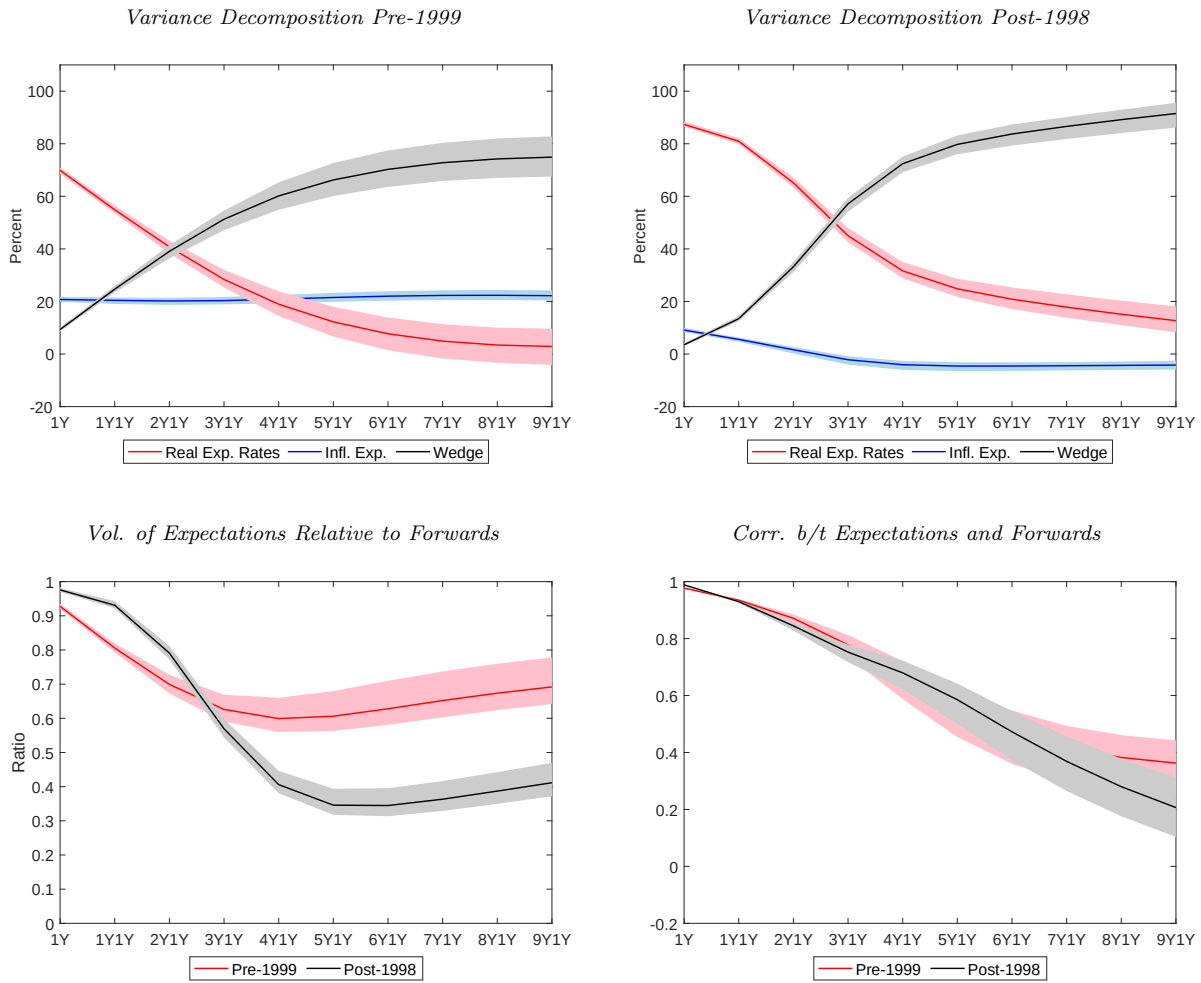


Figure SA1-3. Decomposing Forward Rates (5-Year Changes). The top panels show the variance decomposition of the 60-month changes of forward rates at different maturities in real expected rates (red line), inflation expectations (blue line) and the wedge (black line) components. The bottom left panel displays the volatility of 60-month changes in short rate expectations relative to forwards for the pre-1999 (red line) and post-1998 (black line) sample. Solid lines denote posterior medians while shaded regions denote the range of 1,000 posterior draws. The sample period is 1983m1–2019m12.

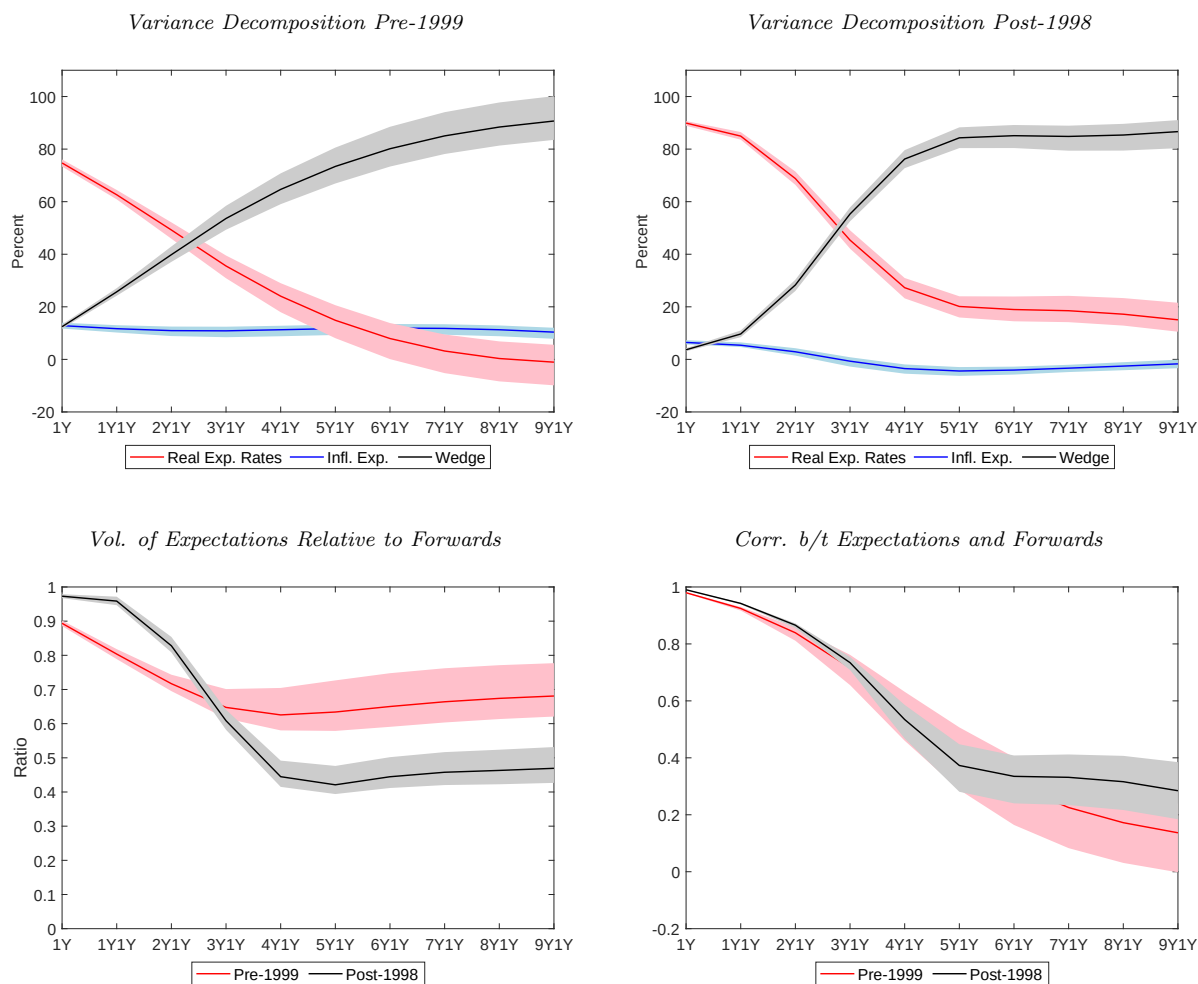
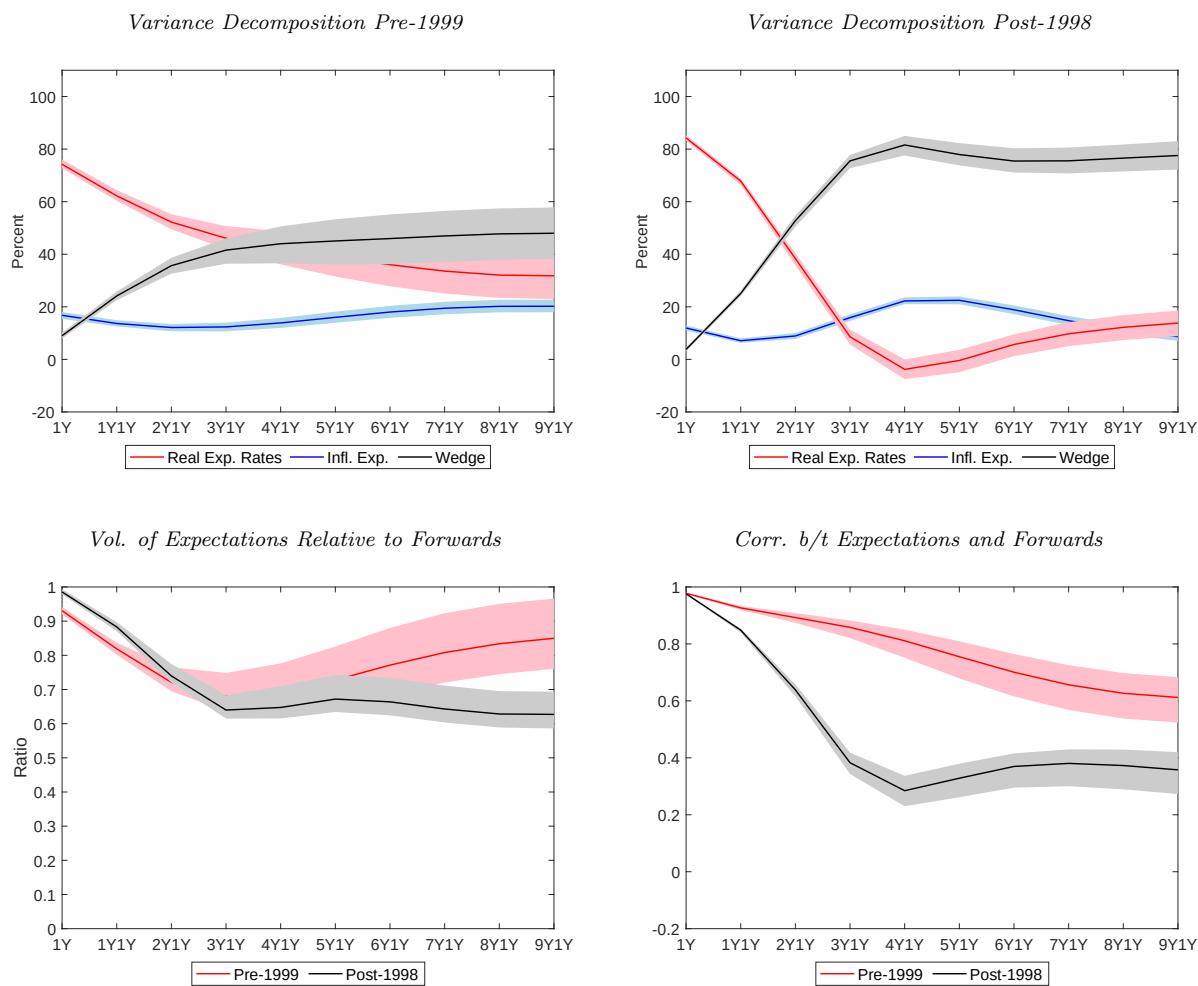


Figure SA1-4. Decomposing Forward Rates (10-Year Changes). The top panels show the variance decomposition of the 120-month changes of forward rates at different maturities in real expected rates (red line), inflation expectations (blue line) and the wedge (black line) components. The bottom left panel displays the volatility of 120-month changes in short rate expectations relative to forwards for the pre-1999 (red line) and post-1998 (black line) sample. Solid lines denote posterior medians while shaded regions denote the range of 1,000 posterior draws. The sample period is 1983m1–2019m12.



SA1-1.5 Spanning Tests: Quadratic Specification

In this section, we generalize the linear specification used in Section 3.5 to a quadratic specification. Specifically, along with the linear terms (as given by equations (3.14)–(3.15)) we include quadratic terms and the interactions between all variables. This can be interpreted as a nonparametric series estimator for the unknown conditional expectation (e.g., [Chen 2007](#)). In practice, this results in 27 explanatory variables (including the constant). Table SA1-1 presents the raw R^2 values from this regression using either forward rates, $\text{fwd}_t(n, m)$, or the survey wedges, $\text{sw}_t(n, m)$, as the dependent variable.

Since our attention is on the behavior of the wedge we will focus on the lower panel of Table SA1-1. The results show that allowing for nonlinearities does meaningfully increase the R^2 . However, the R^2 values are still far from fully explaining the movements in the wedges. For example, for 1-year and 10-year changes, the explained variation is, at most, between 59% and 66%. At 5-year changes, the explained variation rises, but never explains even 77% and is often substantially lower (across all 1,000 posterior draws). Taken in sum, even by taking the most conservative approach of allowing for many regressors, without regularization, and using raw R^2 values, we still conclude that there are important additional drivers of the wedges which are not spanned by the factors which drive short-term interest rate expectations.

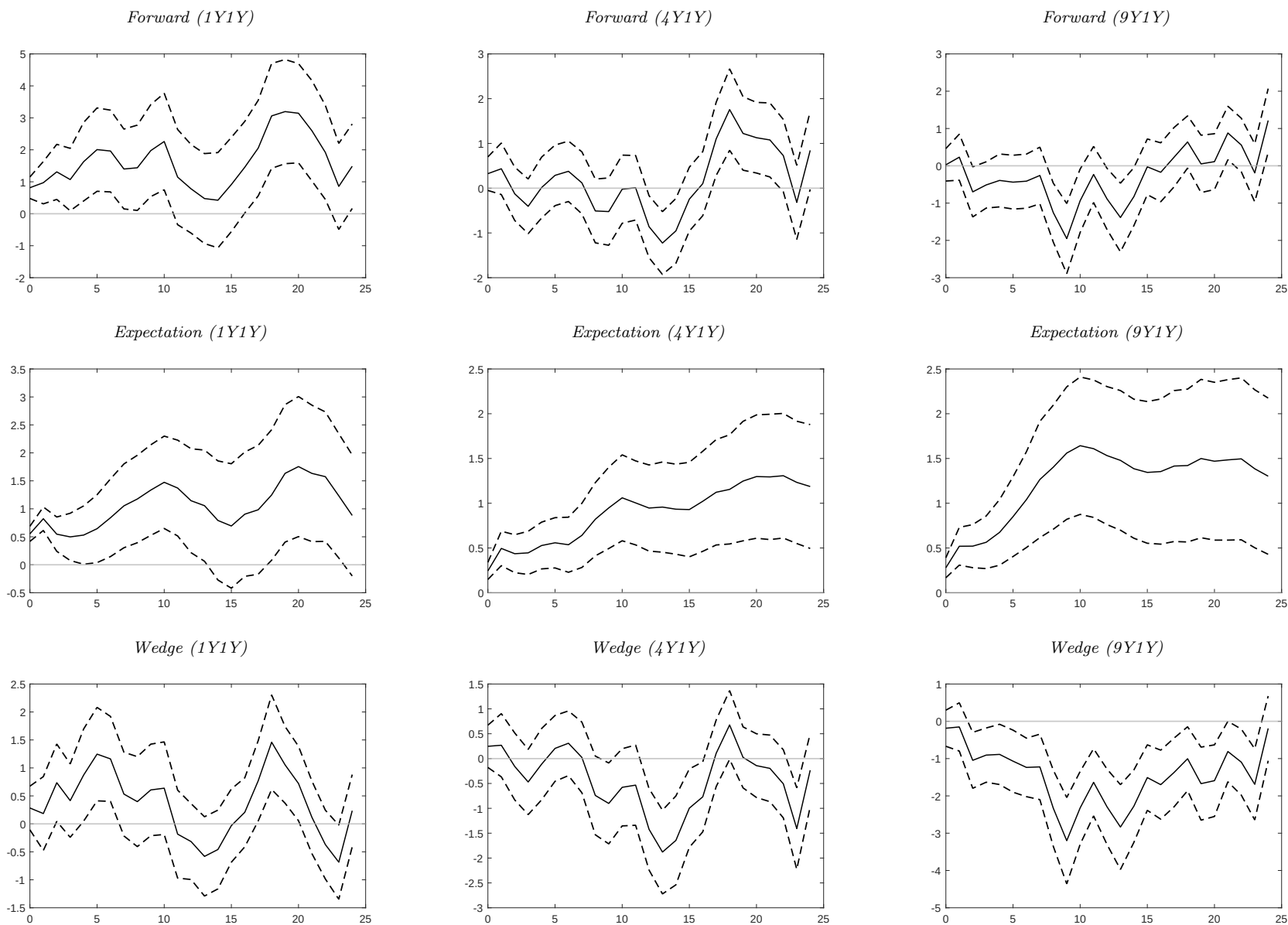
Table SA1-1. Spanning Tests. The top panel (bottom panel) presents R^2 from regressions of changes in the forward rate (wedge) on a constant and all interactions of the six state variables, $\hat{x}_{t|t}$ and $\hat{\omega}_{t|t}$ (27 regressors in all). Minimum and maximum statistics are taken across 1,000 posterior draws. The sample period is 1983m1–2019m12.

Forward Rates: $\text{fwd}_t(n, m)$										
	1Y	1Y1Y	2Y1Y	3Y1Y	4Y1Y	5Y1Y	6Y1Y	7Y1Y	8Y1Y	9Y1Y
<i>12-month Changes</i>										
Min. R^2 (%)	94.1	77.6	64.8	56.7	51.6	48.0	46.1	44.9	43.9	43.1
Max. R^2 (%)	95.3	82.9	72.6	66.5	62.5	60.3	59.5	59.2	59.2	59.3
<i>60-month Changes</i>										
Min. R^2 (%)	98.4	93.8	87.5	79.8	73.1	68.6	66.3	65.4	65.0	64.9
Max. R^2 (%)	98.7	95.0	90.3	84.9	80.7	78.0	76.5	76.1	75.8	75.5
<i>120-month Changes</i>										
Min. R^2 (%)	97.7	90.1	83.9	79.7	77.3	76.0	74.7	73.4	72.2	70.9
Max. R^2 (%)	98.1	91.9	86.5	83.2	81.9	81.0	80.1	79.3	78.4	77.4
Survey Wedges: $\text{sw}_t(n, m)$										
	1Y	1Y1Y	2Y1Y	3Y1Y	4Y1Y	5Y1Y	6Y1Y	7Y1Y	8Y1Y	9Y1Y
<i>12-month Changes</i>										
Min. R^2 (%)	23.3	26.1	27.9	31.9	36.1	39.7	42.7	44.9	46.0	46.6
Max. R^2 (%)	37.9	41.0	42.1	44.5	48.7	52.6	55.8	57.8	59.1	59.9
<i>60-month Changes</i>										
Min. R^2 (%)	43.0	50.8	52.7	55.0	58.7	61.1	62.7	64.2	65.7	67.1
Max. R^2 (%)	52.1	60.4	63.1	66.2	69.5	72.0	73.5	74.4	75.7	76.8
<i>120-month Changes</i>										
Min. R^2 (%)	53.6	55.5	57.9	58.1	56.7	54.4	53.6	53.9	54.1	53.7
Max. R^2 (%)	61.5	63.7	65.4	65.7	65.2	64.9	64.5	64.6	64.7	64.1

SA1-1.6 Conditional Properties

In this section, we provide the estimated impulse response functions using the non-orthogonalized monetary policy shocks from [Bauer and Swanson \(2023\)](#). Figure [SA1-5](#) presents these impulse response functions and is the counterpart to Figure [8](#) in the main text. The results using the non-orthogonalized shock are similar to those in Figure [8](#) except we observe a starker difference in the relative behavior of expectations as compared to wedges. In Figure [SA1-5](#), at all maturities, expected short rates rise more firmly; furthermore, the wedge at the 4Y1Y and 9Y1Y maturities, declines more strongly.

Figure SA1-5. Impulse Response Functions: Forwards and Expectations (Non-Orthogonalized Shock). This figure presents impulse responses to a monetary policy shock of forwards, expectations, and wedges for selected forward maturities using the local projection model described by equation (3.13). The coefficient of interest is estimated using instrumental variables with the non-orthogonalized shock of [Bauer and Swanson \(2023\)](#) as an instrument for the two-year yield. The sample period is 1998m2–2019m12.



SA1-2 Dynamic NS Model

SA1-2.1 Model Setup

Here we follow [Aruoba \(2020\)](#) who employs a dynamic [Nelson and Siegel \(1987\)](#) (NS) model to fit the survey-based term structure of inflation expectation from US professional forecasters. Let $i_{t+\tau}$ be the true (unobserved) expected average short rate from month t to $t + \tau$ for $\tau > 0$. We assume,

$$i_{t+\tau} = L_t - \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} \right) S_t + \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right) C_t,$$

where L_t, S_t and C_t denote level, slope and curvature factors, respectively, and λ is a parameter that regulates the factor loadings for the entire term structure of expectations.² The expected average short rate from month $t + \tau_1$ to $t + \tau_2$,

$$i_{t+\tau_1 \rightarrow t+\tau_2} = \frac{\tau_2}{\tau_2 - \tau_1} i_{t+\tau_2} - \frac{\tau_1}{\tau_2 - \tau_1} i_{t+\tau_1},$$

which can then be expressed as

$$i_{t+\tau_1 \rightarrow t+\tau_2} = L_t + \frac{e^{-\lambda\tau_1} - e^{-\lambda\tau_2}}{\lambda(\tau_2 - \tau_1)} (C_t - S_t) + \left(\frac{\tau_1 e^{-\lambda\tau_1} - \tau_2 e^{-\lambda\tau_2}}{\tau_2 - \tau_1} \right) C_t,$$

which corresponds to equation (5) of [Aruoba \(2020\)](#). We can rearrange the factors to obtain,

$$\begin{aligned} i_{t+\tau_1 \rightarrow t+\tau_2} &= L_t - \frac{e^{-\lambda\tau_1} - e^{-\lambda\tau_2}}{\lambda(\tau_2 - \tau_1)} S_t \\ &\quad + \frac{1}{\tau_2 - \tau_1} \left[\frac{(1 + \lambda\tau_1) e^{-\lambda\tau_1} - (1 + \lambda\tau_2) e^{-\lambda\tau_2}}{\lambda} \right] C_t. \end{aligned}$$

The model then defines the observation equation mapping survey-based expectations and the three factors. Take, as an example, the two-quarter-ahead expected (quarterly) average of the short-term rate:

$$\begin{aligned} i_{Q+2} &= \frac{1}{3} (i_{t+4} + i_{t+5} + i_{t+6}) \\ &= \frac{1}{3} i_{t+3 \rightarrow t+4} + \frac{1}{3} i_{t+4 \rightarrow t+5} + \frac{1}{3} i_{t+5 \rightarrow t+6}. \\ &= i_{t+3 \rightarrow t+6}. \end{aligned}$$

This can then be expressed as,

$$i_{t+3 \rightarrow t+6} = L_t + \frac{e^{-3\lambda} - e^{-6\lambda}}{3\lambda} (C_t - S_t) + \left(\frac{3e^{-3\lambda} - 6e^{-6\lambda}}{3} \right) C_t$$

²It is important to emphasize that this parameter λ has no association with the parameter λ in the main text. We keep this notation as it aligns with that of [Aruoba \(2020\)](#).

$$= L_t - \frac{1}{3} \left(\frac{e^{-3\lambda} - e^{-6\lambda}}{\lambda} \right) S_t + \frac{1}{3} \left[\frac{(1+3\lambda)e^{-3\lambda} - (1+6\lambda)e^{-6\lambda}}{\lambda} \right] C_t.$$

The unobserved level, slope and curvature factors evolve according to a random walk and two (mean-zero) autoregressive processes of order two. The model transition equation is then expressed as

$$\begin{pmatrix} L_t \\ S_t \\ C_t \\ S_{t-1} \\ C_{t-1} \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \rho_{S1} & 0 & \rho_{S2} & 0 \\ 0 & 0 & \rho_{C1} & 0 & \rho_{C2} \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \begin{pmatrix} L_{t-1} \\ S_{t-1} \\ C_{t-1} \\ S_{t-2} \\ C_{t-2} \end{pmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \varepsilon_t,$$

where ε_t includes the innovations to level, slope and curvature. These innovations are i.i.d. Gaussian and mean zero. Consistent with the above, the observation equation is, for each survey,

$$\mathbb{E}^s[i_{\tau_1 \rightarrow \tau_2}] = \begin{bmatrix} 1 & \frac{e^{-\lambda\tau_2} - e^{-\lambda\tau_1}}{\lambda(\tau_2 - \tau_1)} & \frac{(1+\lambda\tau_1)e^{-\lambda\tau_1} - (1+\lambda\tau_2)e^{-\lambda\tau_2}}{(\tau_2 - \tau_1)\lambda} & 0 & 0 \end{bmatrix} \begin{pmatrix} L_t \\ S_t \\ C_t \\ S_{t-1} \\ C_{t-1} \end{pmatrix} + \eta_t^{\tau_1 \rightarrow \tau_2}.$$

The model is then defined by the set of parameters

$$\Theta = \left(\rho_{S1} \quad \rho_{S2} \quad \rho_{C1} \quad \rho_{C2} \quad \sigma_{L,2} \quad \sigma_{L,1} \quad \sigma_S \quad \sigma_C \quad \lambda \right)'.$$

Consistent with baseline model, we allow a break in the volatility of the level factor after 1999. This results in two parameters: $\sigma_{L,1}$ and $\sigma_{L,2}$. In addition to the above, we have four parameters governing the variance of the measurement errors to the survey data, which restricts the behavior of $\eta_t^{\tau_1 \rightarrow \tau_2}$ across responses. These measurement errors are modeled in the same way as in the baseline model in the main text. As with the baseline model, this specification is estimated with Bayesian methods. The priors are set as follows. The standard deviations, $\sigma_{L,1}$ and $\sigma_{L,2}$, each have a prior distributed according to a beta distribution with mean 0.1 and standard deviation of 0.03. As in the case of the baseline model the level factor is assumed to be slow-moving relative to the other two factors. The standard deviations of the innovations in the slope and curvature factors (σ_S, σ_C) have looser priors. They are distributed according to an inverse gamma distribution with mean 0.2 and standard deviation of 2. We use the same prior for the parameter λ . The parameters ρ_{S1}, ρ_{C1} are normally distributed with mean 1.25 and standard deviation of 1. The parameters ρ_{S2}, ρ_{C2} are normally distributed with mean -0.3 and standard deviation of 1. The priors are also fairly loose. In terms of their posterior distribution, the parameter λ is tightly estimated in the range $0.070 - 0.072$ (i.e. the 90% of posterior draws). Consistent with the baseline model, the standard deviation of the level factor drops from the range $0.12 - 0.15$ before 1999 to the range $0.07 - 0.09$ afterwards.

Figure SA1-6 presents scatterplots comparing the realized survey data to the fit of the baseline model and the fit of the dynamic NS model. Both scatterplots are tightly distributed around the 45 degree line. The root mean-square error (RMSE) for fitting the survey forecasts of the nominal short rate for the baseline model is 0.1376. The corresponding RMSE for the dynamic NS model is 0.1453. Figure SA1-7 is the counterpart of Figure 2 in the main text for the NS model. Comparing these figures we can observe that the dynamic NS model also fits the survey forecasts very well. Consequently, the wedge between observed forwards and model-based expectations from the dynamic NS model or the baseline model of Section 2.3 will be very similar. This is shown in Figure SA1-8 which compares these two different measures of the survey wedge for selected forward maturities. As the charts show, the two measures are very close in magnitude and share common dynamic behavior. In subsequent sections, we replicate the key results in the paper using this alternative fitted model.

Figure SA1-6. Model Fit Scatterplots. This figure provides scatterplots comparing the realized survey forecasts to the associated model fit. The left chart presents results for the baseline model and the right chart presents results for the dynamic NS model. The sample period is 1983m1–2019m12.

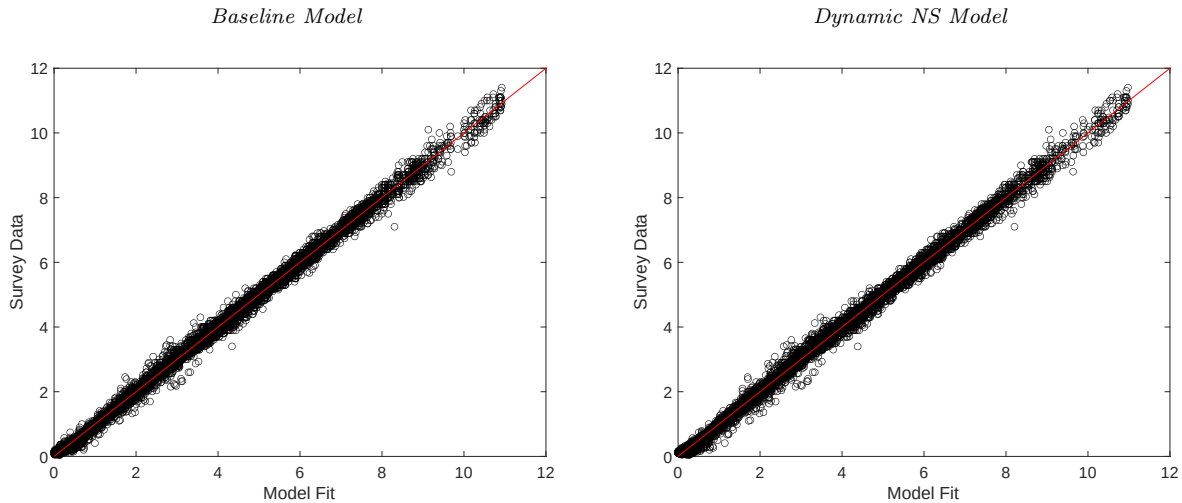


Figure SA1-7. Term Structure of Interest Rate Expectations. This figure compares model-implied survey forecasts based on the dynamic NS model to selected individual survey-horizon pairs. The top panel displays four-quarter ahead forecasts taken in the last month of the quarter (“Short-Term”); the middle panel displays forecasts for the one-year average interest rate three years ahead taken in the first half of the year (“Medium-Term”); the bottom panel displays forecasts for the one-year average interest rate five years ahead taken in the first half of the year (“Long-Term”). The dots indicate forecasts from different surveys: BCEI and BCFF shown in red, CE shown in blue; EF and SPF are shown in green in the top and middle panel, respectively. The black line denotes the median prediction from the model, and the dark grey area the 99% posterior coverage interval. Light grey shaded areas denote NBER recessions. The sample period is 1983m1–2019m12.

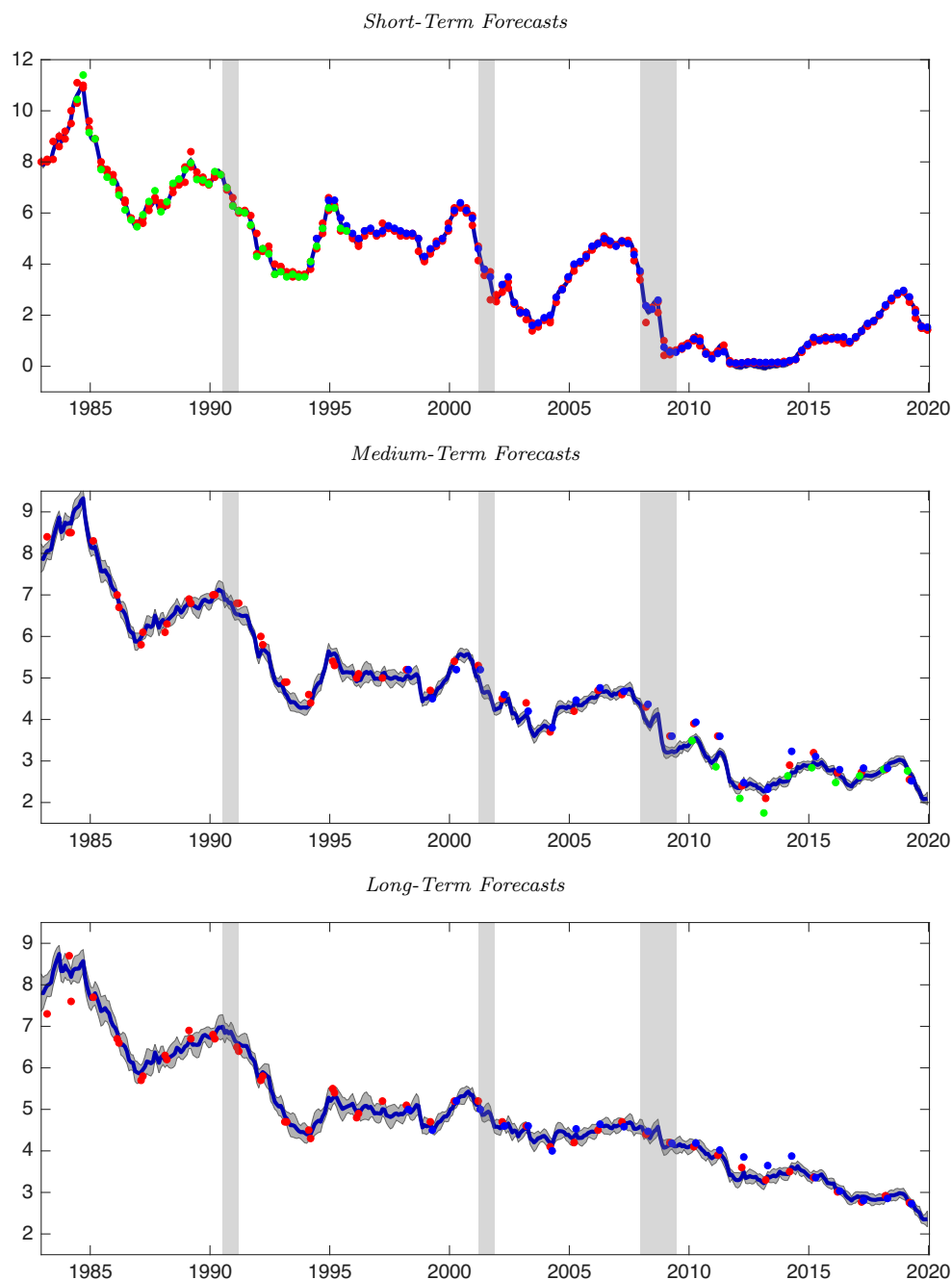
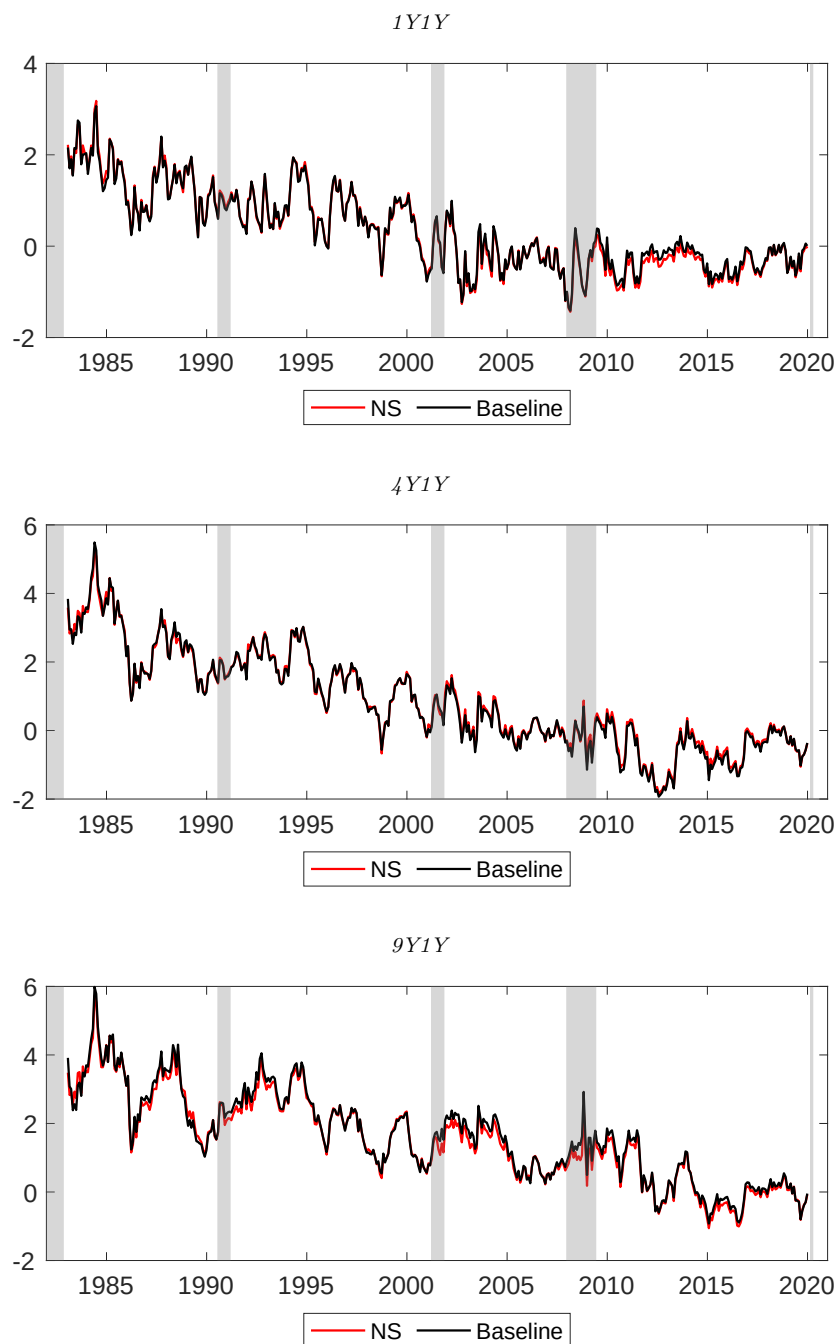


Figure SA1-8. Comparing Model-Based Survey Wedges. This figure compares model-implied survey wedges based on the expectations formation model introduced in Section 2.3 (labelled “Baseline”) against those of the dynamic NS model for selected maturities. The sample period is 1983m1–2019m12.



SA1-2.2 Testing the EH With Survey-Based Expectations

Table SA1-2 provides the corresponding results to Table 3 using the dynamic NS model to construct the term structure of expectations. Comparing the two tables, we can observe that the outcomes of the new test of the EH are very similar based on either model. Across all of the forward maturities, the full sample, and the two subsamples, the EH is almost uniformly rejected at the 1% level. The only exception is the 1Y maturity where the maximum test statistic across posterior draws comfortably rejects at the 5% level. Nonetheless, as in Table 3, the EH is strongly rejected at conventional significance levels.

Table SA1-2. Tests of the Expectations Hypothesis (Model-Based). This table reports results of the Elliott and Müller (2006) test applied to the null hypothesis described by equation (3.6) using the dynamic NS model-based short rate expectations for selected forward maturities. For reference, the test rejects for small values of the test statistic with 1% critical value -11.05 and 5% critical value -8.36 . The minimum and maximum values of the test statistic across 1,000 posterior draws are reported. The sample period is 1983m1–2019m12.

	1Y	1Y1Y	2Y1Y	3Y1Y	4Y1Y	5Y1Y	6Y1Y	7Y1Y	8Y1Y	9Y1Y
<i>Full Sample</i>										
Max	-40.7	-48.1	-50.1	-50.0	-49.4	-48.7	-47.8	-47.1	-46.6	-46.3
Min	-42.2	-48.8	-50.9	-51.1	-50.7	-50.1	-49.5	-48.9	-48.4	-48.1
<i>1983–1998</i>										
Max	-17.4	-19.2	-21.5	-21.9	-21.5	-20.7	-19.8	-18.8	-17.9	-17.2
Min	-19.2	-20.9	-23.3	-23.9	-23.8	-23.3	-22.5	-21.7	-20.9	-20.3
<i>1999–2019</i>										
Max	-10.3	-14.9	-21.7	-23.5	-23.8	-24.2	-24.9	-25.7	-26.5	-27.1
Min	-11.8	-16.5	-23.2	-25.3	-25.8	-26.4	-27.1	-27.9	-28.6	-29.1

SA1-2.3 Variance Decompositions: 1-Year, 2-Year, 5-Year and 10-Year Changes

In this section, we provide variance decompositions for 1-year, 2-year, 5-year and 10-year changes using the dynamic NS model to construct the term structure of expectations. In particular, we provide the corresponding variance decompositions in Figures SA1-9, SA1-10, SA1-11, and SA1-12, respectively.

Figure SA1-9. Decomposing Forward Rates (1-Year Changes). The top panels show the variance decomposition of the 12-month changes of forward rates at different maturities in expected rates (red line) and the wedge (black line) components based on the dynamic NS model. The bottom left panel displays the volatility of 12-month changes in short rate expectations relative to forwards for the pre-1999 (red line) and post-1998 (black line) sample. Solid lines denote posterior medians while shaded regions denote the range of 1,000 posterior draws. The sample period is 1983m1–2019m12.

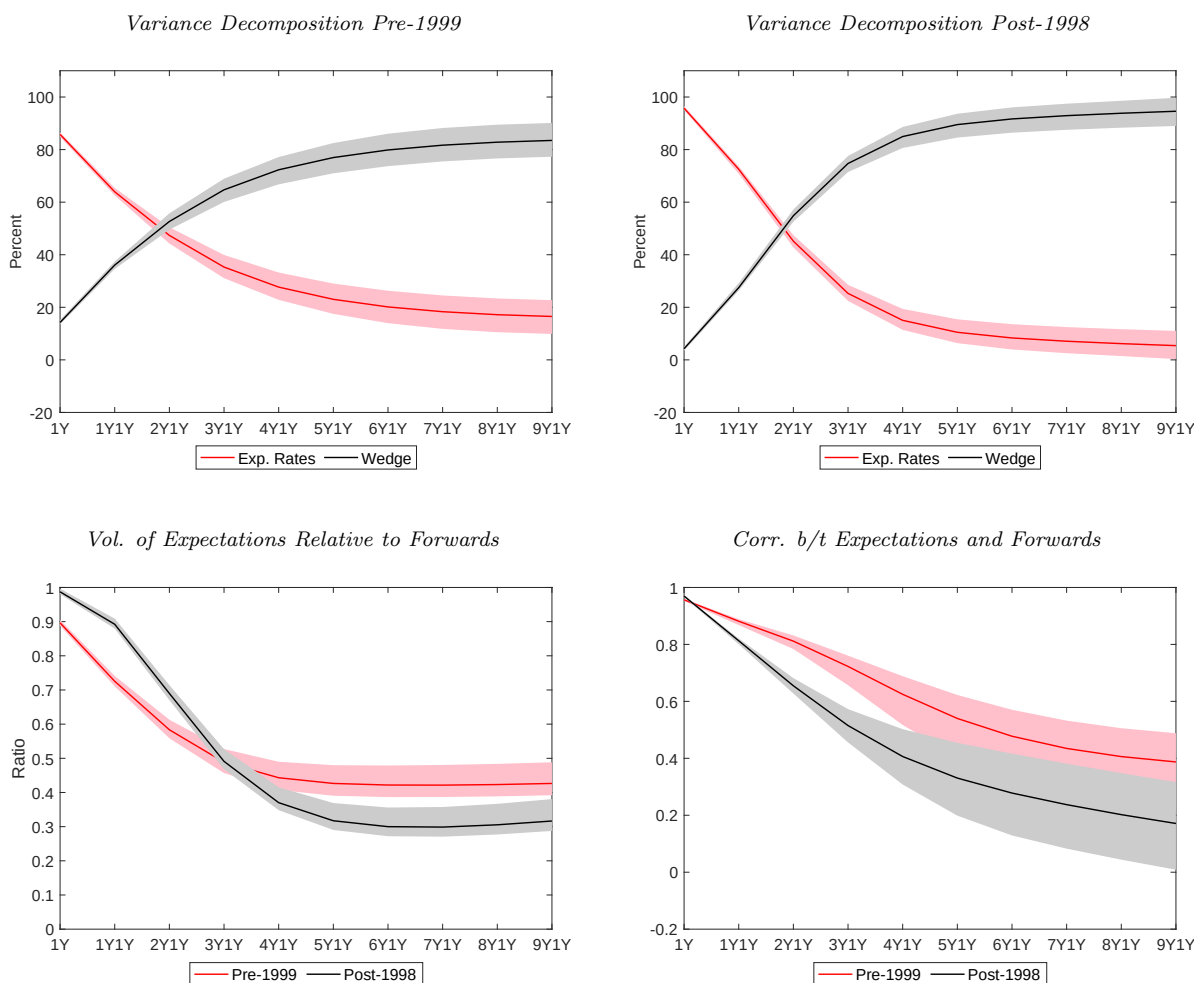


Figure SA1-10. Decomposing Forward Rates (2-Year Changes). The top panels show the variance decomposition of the 24-month changes of forward rates at different maturities in expected rates (red line) and the wedge (black line) components based on the dynamic NS model. The bottom left panel displays the volatility of 24-month changes in short rate expectations relative to forwards for the pre-1999 (red line) and post-1998 (black line) sample. Solid lines denote posterior medians while shaded regions denote the range of 1,000 posterior draws. The sample period is 1983m1–2019m12.

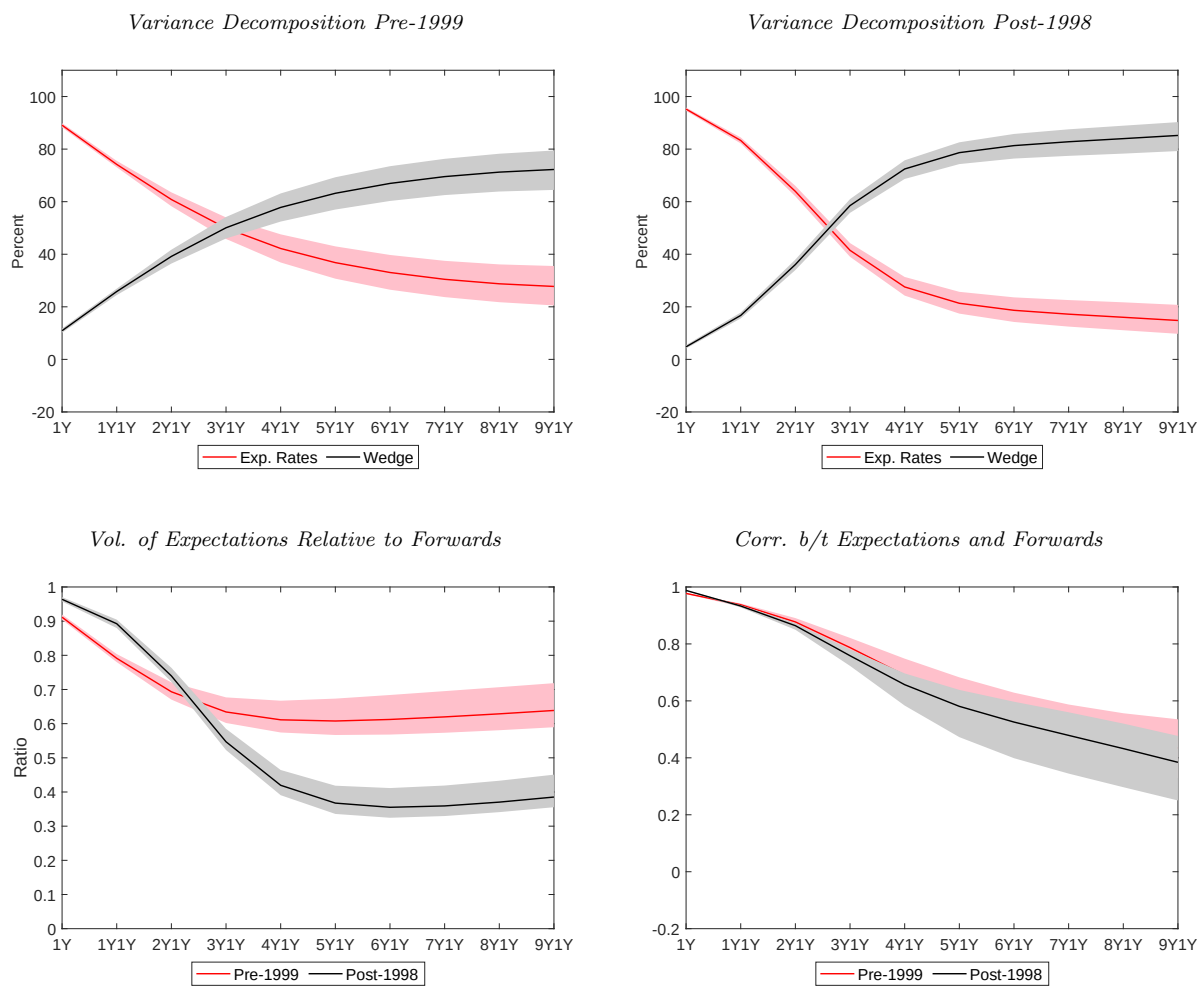


Figure SA1-11. Decomposing Forward Rates (5-Year Changes). The top panels show the variance decomposition of the 60-month changes of forward rates at different maturities in expected rates (red line) and the wedge (black line) components based on the dynamic NS model. The bottom left panel displays the volatility of 60-month changes in short rate expectations relative to forwards for the pre-1999 (red line) and post-1998 (black line) sample. Solid lines denote posterior medians while shaded regions denote the range of 1,000 posterior draws. The sample period is 1983m1–2019m12.

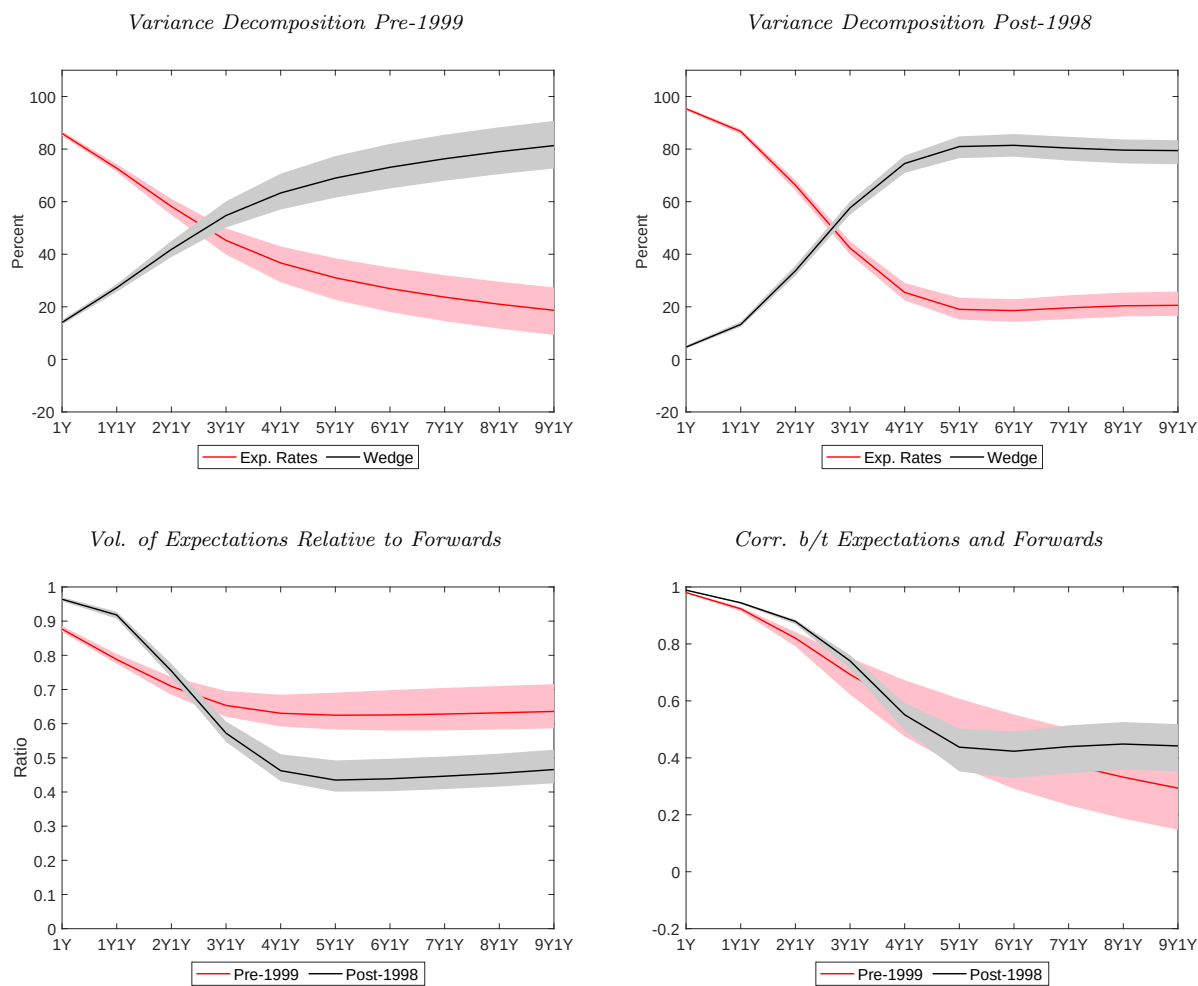
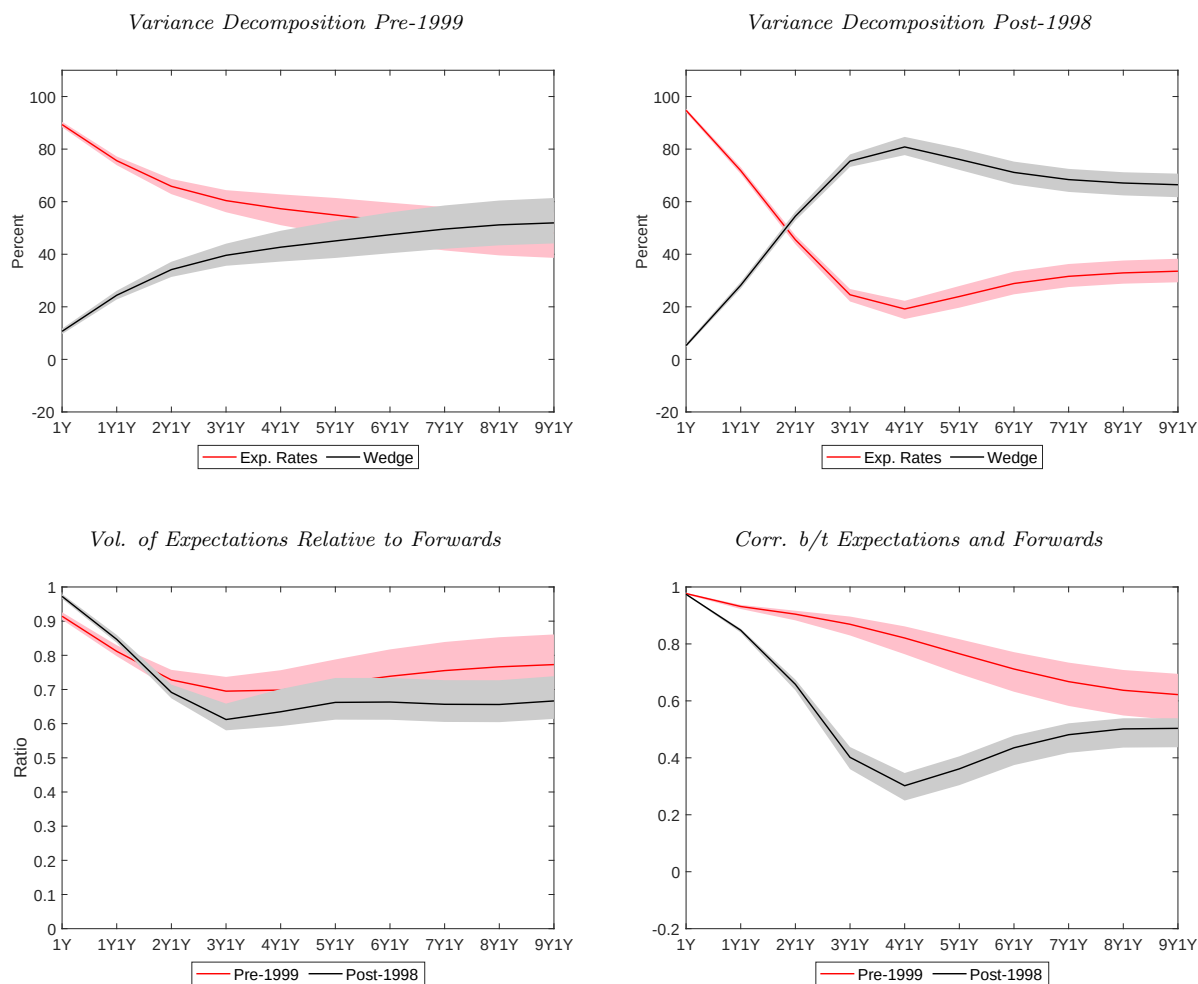


Figure SA1-12. Decomposing Forward Rates (10-Year Changes). The top panels show the variance decomposition of the 120-month changes of forward rates at different maturities in expected rates (red line) and the wedge (black line) components based on the dynamic NS model. The bottom left panel displays the volatility of 120-month changes in short rate expectations relative to forwards for the pre-1999 (red line) and post-1998 (black line) sample. Solid lines denote posterior medians while shaded regions denote the range of 1,000 posterior draws. The sample period is 1983m1–2019m12.



SA1-2.4 Conditional Properties

Figures SA1-13 and SA1-14 provide the corresponding impulse responses to Figure 8 and Figure SA1-5, respectively, using the dynamic NS model to construct the term structure of expectations. The estimated impulse response functions are very similar using either the expectations formation model introduced in Section 2.3 or the dynamic NS model.

Figure SA1-13. Impulse Response Functions: Forwards and Expectations (Dynamic NS Model). This figure presents impulse responses to a monetary policy shock of forwards, expectations, and wedges for selected forward maturities using the local projection model described by equation (3.13). The coefficient of interest is estimated using instrumental variables with the orthogonalized shock of [Bauer and Swanson \(2023\)](#) as an instrument for the two-year yield. The sample period is 1998m2–2019m12.

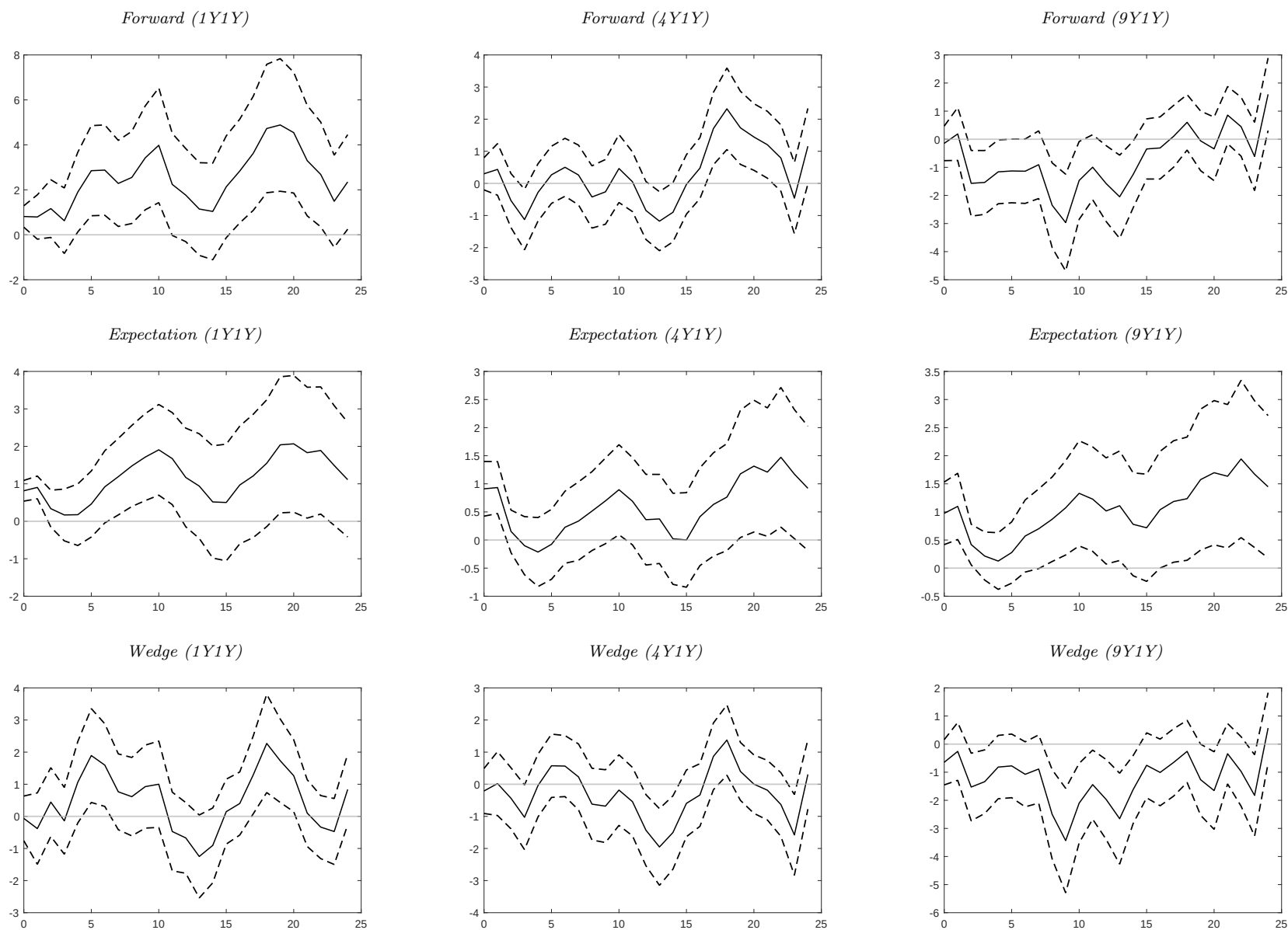
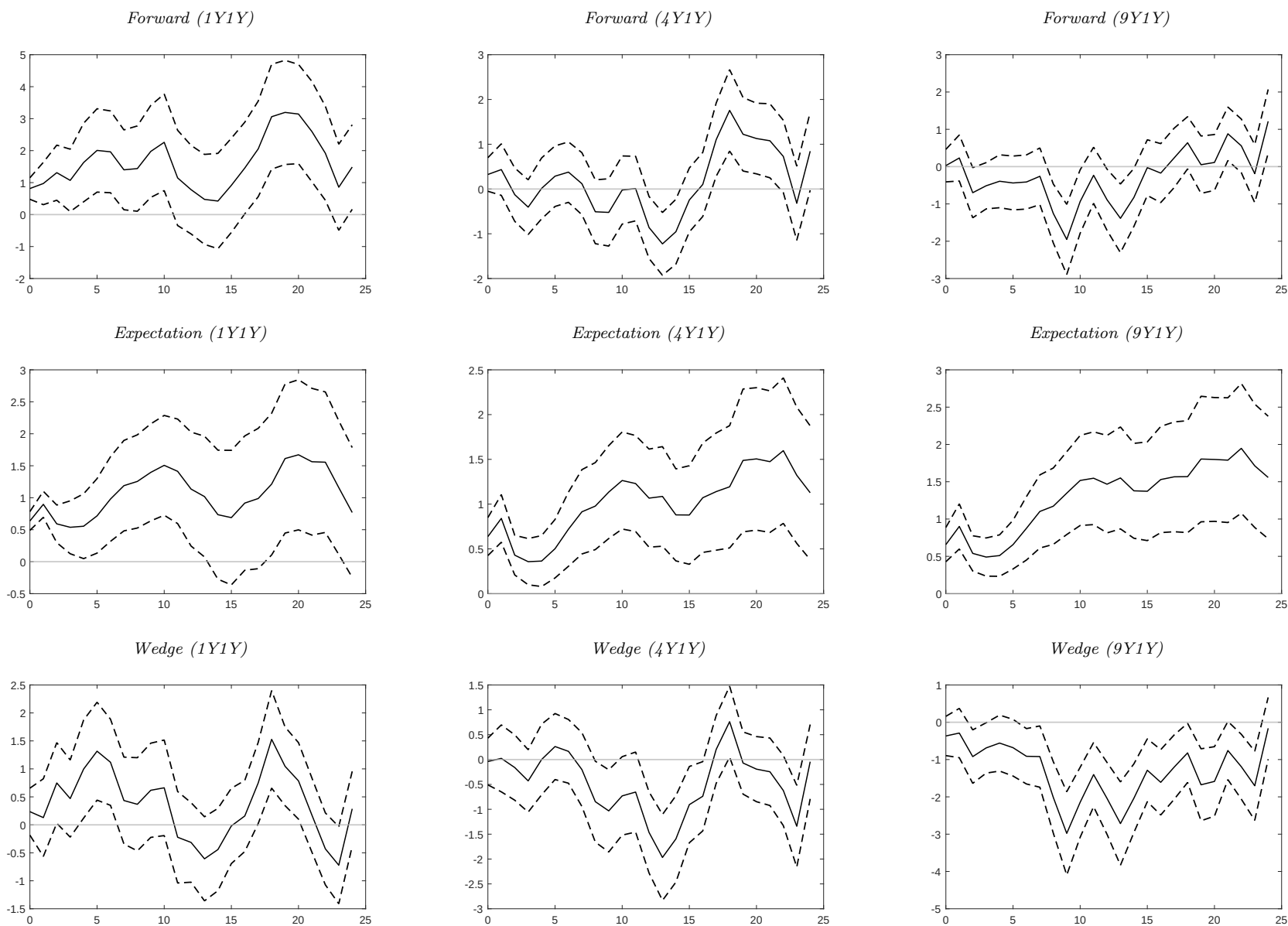


Figure SA1-14. Impulse Response Functions: Forwards and Expectations (Dynamic NS Model; Non-Orthogonalized Shock). This figure presents impulse responses to a monetary policy shock of forwards, expectations, and wedges for selected forward maturities using the local projection model described by equation (3.13). The coefficient of interest is estimated using instrumental variables with the non-orthogonalized shock of [Bauer and Swanson \(2023\)](#) as an instrument for the two-year yield. The sample period is 1998m2–2019m12.



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